

Human-driven trophic simplification in Azorean lake ecosystems

About this script. This R Markdown reproduces the analyses for the manuscript on trophic structure change in Azorean lakes over the past ~2000 years. It loads `data/clean_source_data_files.RData` (raw species matrices and functional-group tables used to recompute ordinations), then loads CSV tables (climate, pollen, guild summaries), runs variance partitioning (phase-based and moving window), sensitivity analyses (phase boundaries, leave-one-out, null models), and produces main and supplementary figures. Custom helper functions are in `functions/source_custom_functions.R`. Run chunks in order; all packages are loaded in the first code section.

```
# Set-up chunk
knitr::opts_chunk$set(
  echo = FALSE,
  warning = FALSE,
  fig.pos = "H",
  fig.width = 12
)
```

Load libraries

Load data

Analysis - Detrended Correspondence Analysis

DCA Diatoms

DCA Chironomids

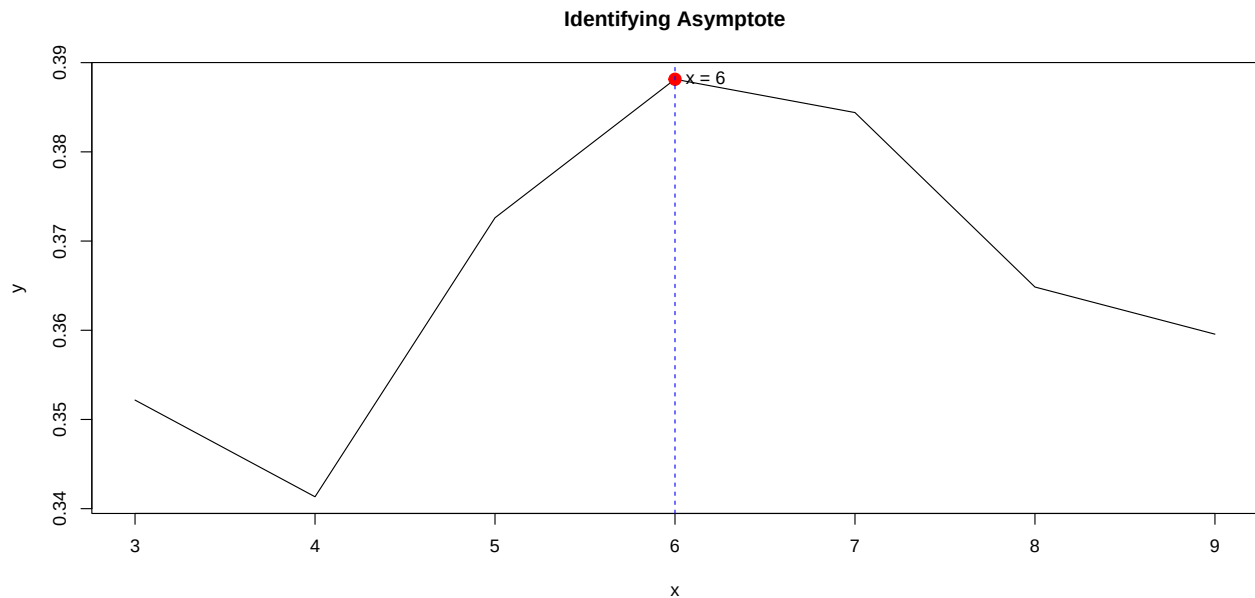
DCA Guilds

Combine DCAs

Stratiplots — top taxa by lake (diatoms & chironomids)

Analysis - Community Trophic Structure Analysis

Get clusters

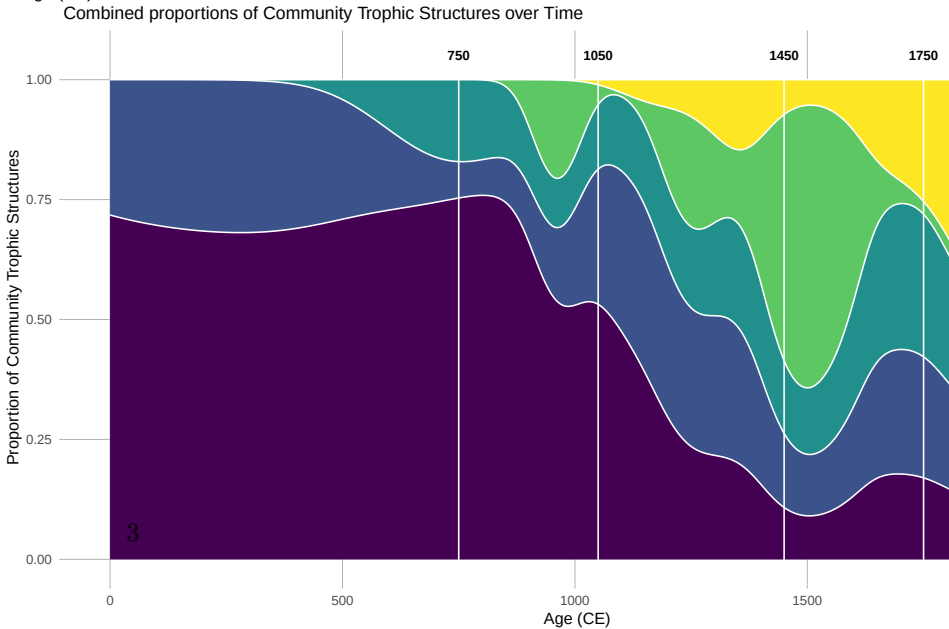
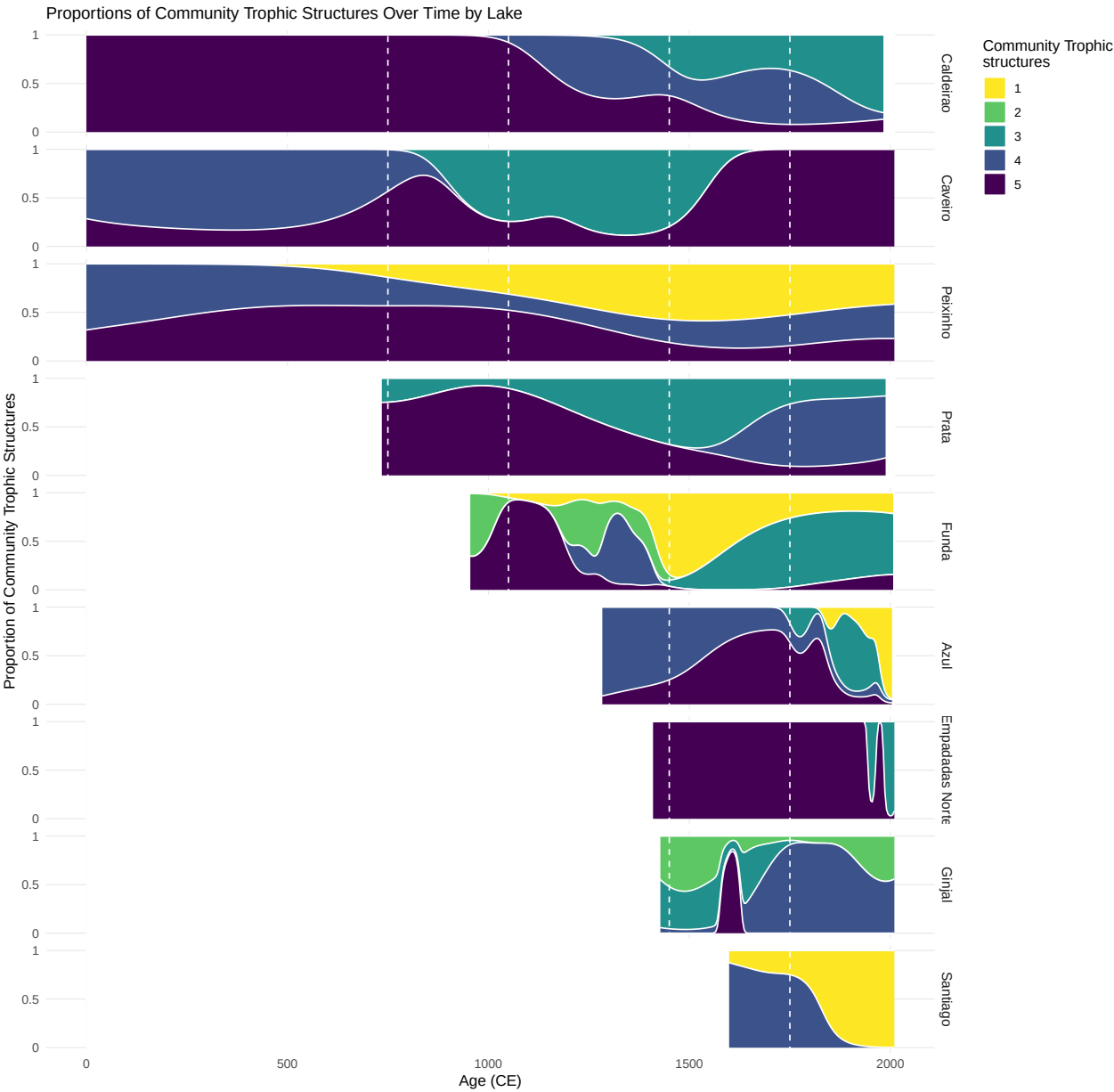


Diversity by community trophic clusters

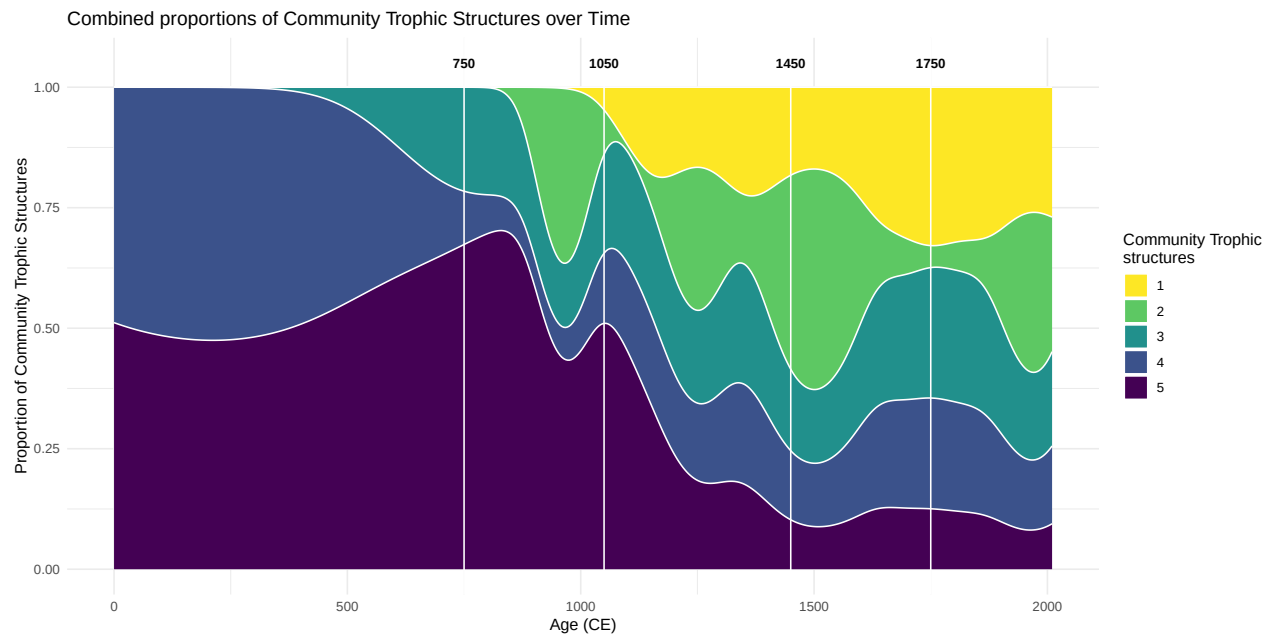
```
## Scale for fill is already present.
```

```
## Adding another scale for fill, which will replace the existing scale.
```

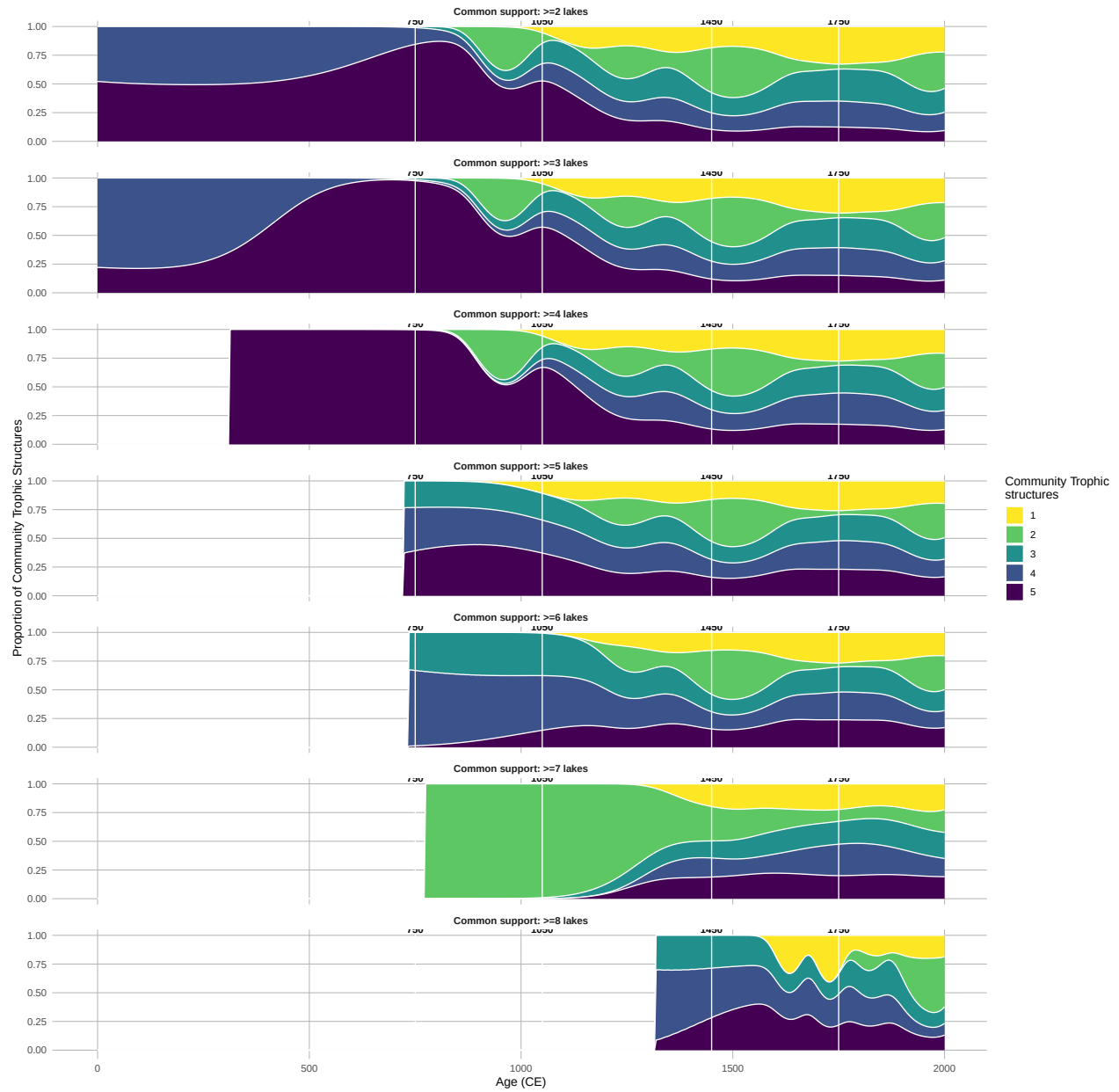
Lake-scale proportions trophic structures



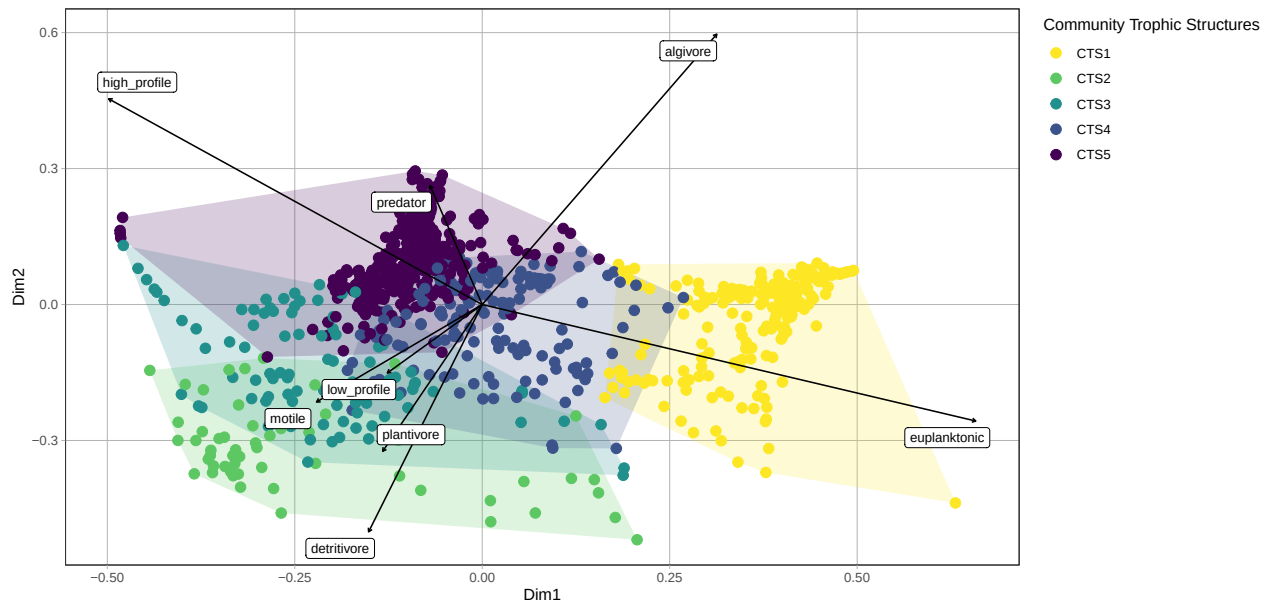
Regional proportions trophic structures



Regional trophic-structure proportions under common-support filtering (0–2000 CE)



Ordination with trophic structures



Random Forest

```
##
## CTS1 CTS2 CTS3 CTS4 CTS5
## 195 55 93 154 411

##
##          CTS1 CTS2 CTS3 CTS4 CTS5
## CTS1    99    0    0    1    0
## CTS2     2   98    0    0    0
## CTS3     0    0   96    0    4
## CTS4     1    0    1   97    1
## CTS5     0    0    0    2   98

## [1] "Classification accuracy: 98 %"

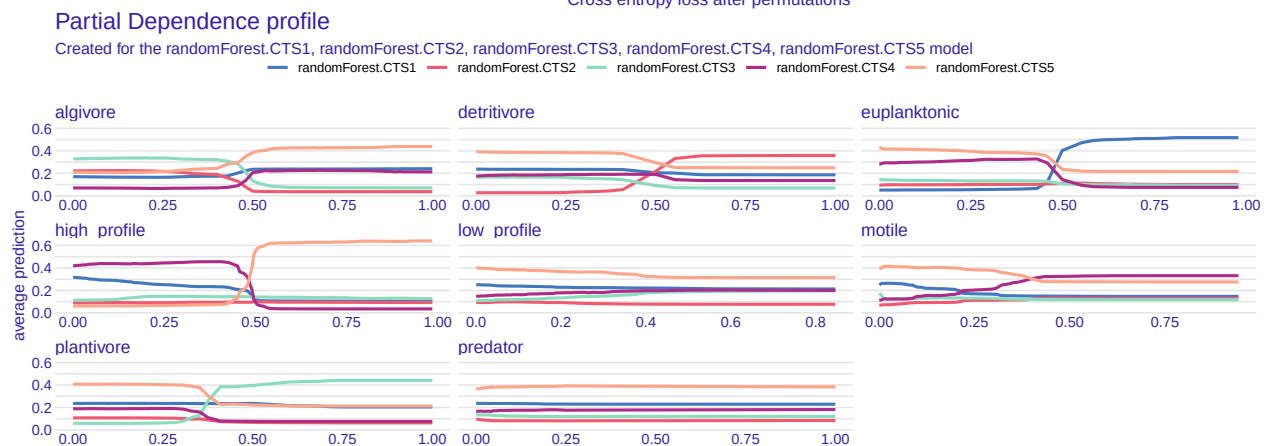
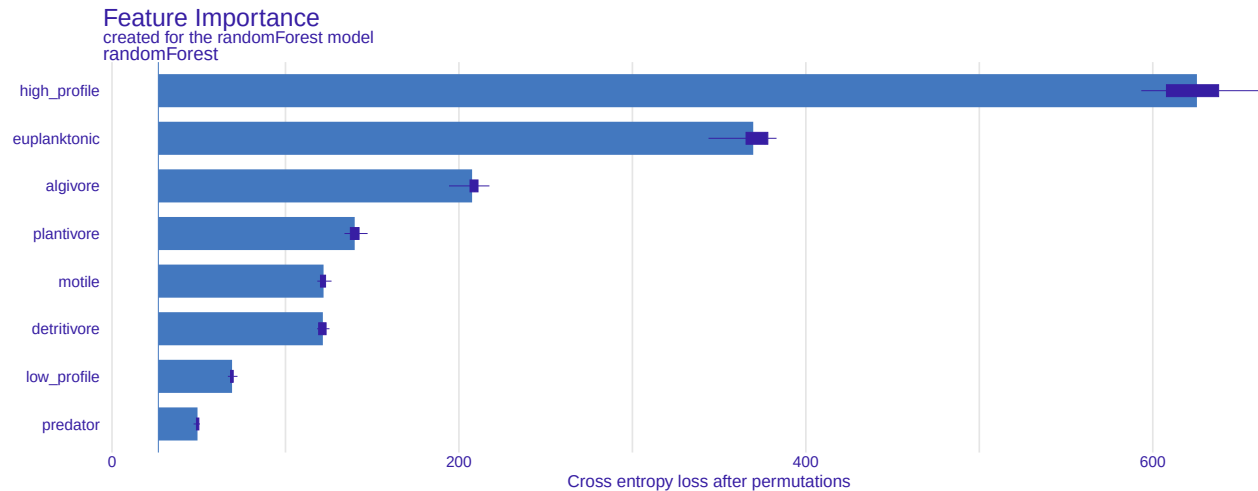
## [1] 0.98

## CTS1 CTS2 CTS3 CTS4 CTS5
## 99 98 96 97 98

## Preparation of a new explainer is initiated
## -> model label      : randomForest ( default )
## -> data             : 908 rows 8 cols
## -> target variable  : 908 values
## -> predict function : yhat.randomForest will be used ( default )
## -> predicted values : No value for predict function target column. ( default )
## -> model_info       : package randomForest , ver. 4.7.1.2 , task multiclass ( default )
## -> predicted values : predict function returns multiple columns: 5 ( default )
## -> residual function : difference between 1 and probability of true class ( default )
## -> residuals        : numerical, min = 0 , mean = 0.02731828 , max = 0.366
## A new explainer has been created!

##          variable mean_dropout_loss      label
## 1 _full_model_      26.71359 randomForest
## 2      predator      49.28947 randomForest
```

```
## 3 low_profile 69.21135 randomForest
## 4 detritivore 121.55391 randomForest
## 5 motile 121.95546 randomForest
## 6 plantivore 139.90804 randomForest
## 7 algivore 207.59851 randomForest
## 8 euplanktonic 369.66605 randomForest
## 9 high_profile 625.45300 randomForest
## 10 _baseline_ 4560.84095 randomForest
```



```
##
## Classification tree:
## rpart(formula = datRF_paleo$amd_clusts ~ ., data = select(datRF_paleo,
##   -c(pred_paleo, amd_clusts)), method = "class", cp = 0.035)
##
## Variables actually used in tree construction:
## [1] algivore detritivore euplanktonic high_profile
##
## Root node error: 497/908 = 0.54736
##
## n= 908
##
##      CP nsplit rel error  xerror   xstd
## 1 0.380282      0  1.000000 1.00000 0.030179
## 2 0.293763      1  0.619718 0.62374 0.028750
## 3 0.076459      2  0.325956 0.33602 0.023489
```

```
## 4 0.068410      3  0.249497 0.26761 0.021438
## 5 0.056338      4  0.181087 0.22535 0.019937
## 6 0.046278      5  0.124748 0.16097 0.017185
## 7 0.035000      6  0.078471 0.11268 0.014585
```

```
## pdf
## 2
```

Analysis - Hierarchical Generalized Additive Models - HGAMs

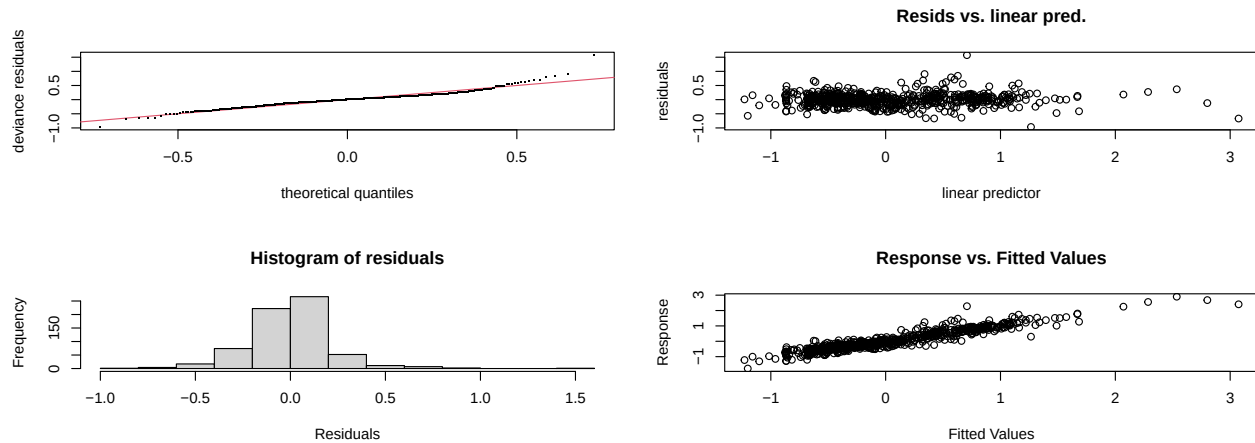
GAM - Diatoms

Regional Smooth

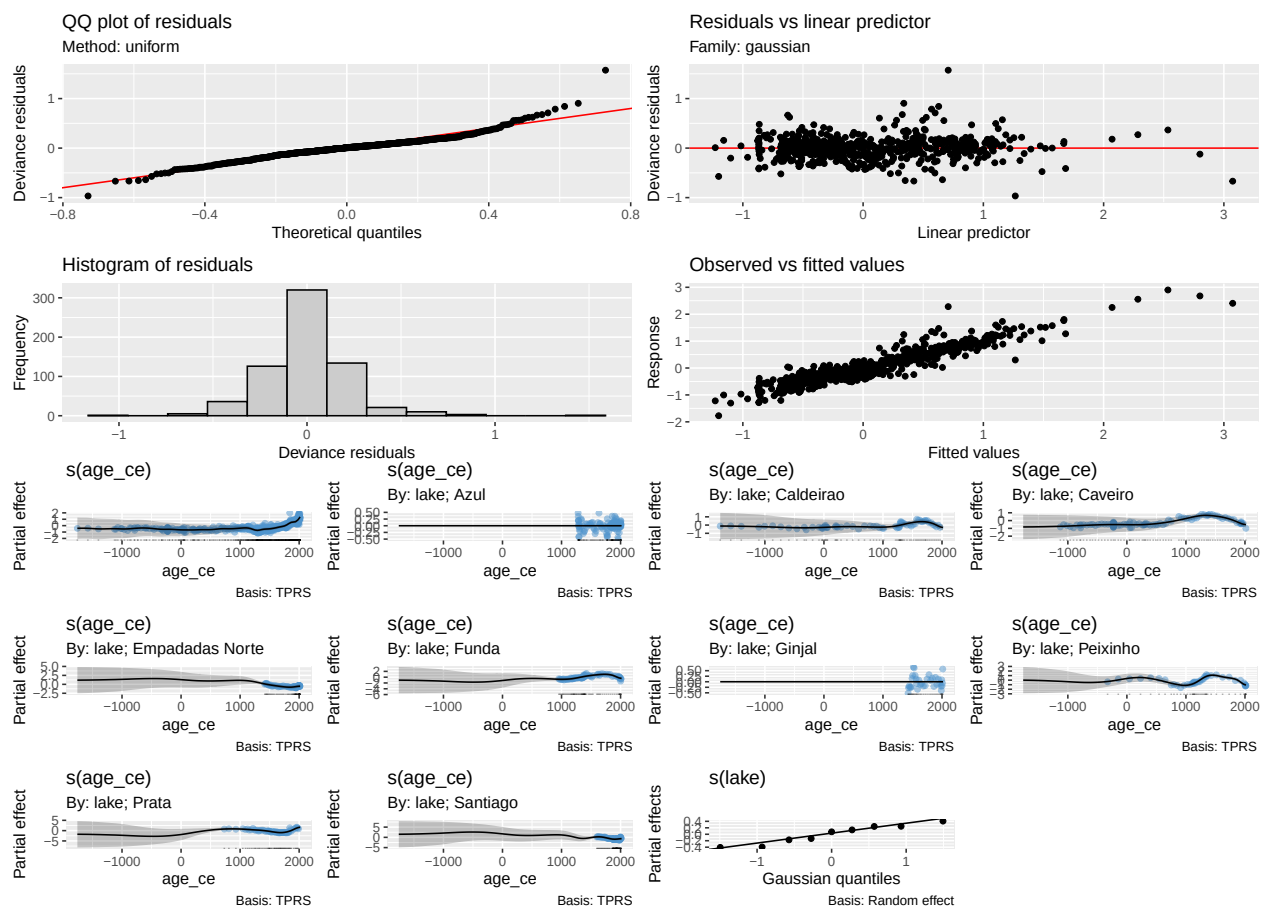
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## DCA1 ~ s(age_ce, bs = "tp", k = 30) + s(age_ce, by = lake, k = 8,
##      m = 1, bs = "tp") + s(lake, bs = "re", k = 8)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.2625      0.1374    1.91  0.0566 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F  p-value
## s(age_ce)      2.314e+01 26.45 42.960 < 2e-16 ***
## s(age_ce):lakeAzul 1.518e-04  7.00  0.000  0.51305
## s(age_ce):lakeCaldeirao 4.462e+00  7.00  7.675 1.13e-05 ***
## s(age_ce):lakeCaveiro 5.084e+00  7.00 16.062 < 2e-16 ***
## s(age_ce):lakeEmpadadas Norte 3.360e+00  7.00 331.314 0.00104 **
## s(age_ce):lakeFundada 4.893e+00  7.00 43.836 0.00434 **
## s(age_ce):lakeGinjal 2.943e-06  7.00  0.000  0.72697
## s(age_ce):lakePeixinho 6.351e+00  7.00 38.216 < 2e-16 ***
## s(age_ce):lakePrata 5.451e+00  7.00 75.352 0.03832 *
## s(age_ce):lakeSantiago 3.452e+00  7.00 253.262 0.02409 *
## s(lake)        5.898e+00  8.00  9.515 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) = 0.881  Deviance explained = 89.2%
## -REML = 99.292  Scale est. = 0.05288  n = 657
##
##              k'      edf  k-index p-value
## s(age_ce)      29 2.313957e+01 1.031452 0.7975
## s(age_ce):lakeAzul 7 1.517915e-04 1.031452 0.7725
## s(age_ce):lakeCaldeirao 7 4.462293e+00 1.031452 0.7700
## s(age_ce):lakeCaveiro 7 5.083585e+00 1.031452 0.7775
## s(age_ce):lakeEmpadadas Norte 7 3.359771e+00 1.031452 0.7950
## s(age_ce):lakeFundada 7 4.893398e+00 1.031452 0.7700
## s(age_ce):lakeGinjal 7 2.942964e-06 1.031452 0.7675
```



```
## s(age_ce):lakePeixinho      7 6.351217e+00 1.031452 0.7450
## s(age_ce):lakePrata         7 5.451007e+00 1.031452 0.7900
## s(age_ce):lakeSantiago      7 3.452201e+00 1.031452 0.7925
## s(lake)                     9 5.897854e+00      NA      NA
```



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 17 iterations.
## Gradient range [-3.580949e-05,4.054034e-05]
## (score 99.2917 & scale 0.0528799).
## Hessian positive definite, eigenvalue range [1.03983e-06,328.0314].
## Model rank = 102 / 102
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##          k'      edf k-index p-value
## s(age_ce)      2.90e+01 2.31e+01    1.03    0.73
## s(age_ce):lakeAzul      7.00e+00 1.52e-04    1.03    0.79
## s(age_ce):lakeCaldeirao  7.00e+00 4.46e+00    1.03    0.79
## s(age_ce):lakeCaveiro   7.00e+00 5.08e+00    1.03    0.81
## s(age_ce):lakeEmpadadas Norte 7.00e+00 3.36e+00    1.03    0.71
## s(age_ce):lakeFundada   7.00e+00 4.89e+00    1.03    0.84
## s(age_ce):lakeGinjal    7.00e+00 2.94e-06    1.03    0.79
## s(age_ce):lakePeixinho  7.00e+00 6.35e+00    1.03    0.75
## s(age_ce):lakePrata     7.00e+00 5.45e+00    1.03    0.80
## s(age_ce):lakeSantiago  7.00e+00 3.45e+00    1.03    0.80
## s(lake)         9.00e+00 5.90e+00      NA      NA
```



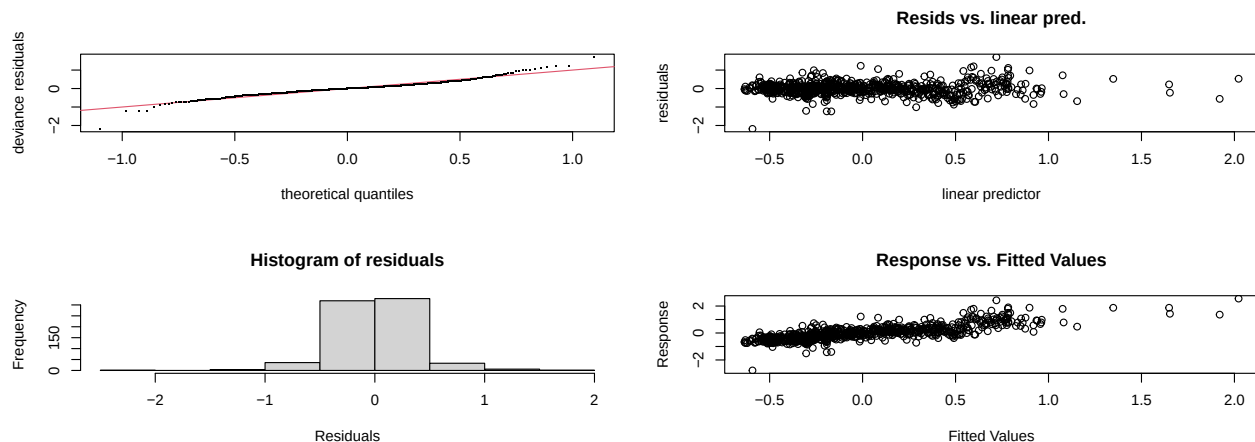
GAM - Chironomids

Regional Smooth

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## DCA1 ~ s(age_ce, bs = "tp", k = 30) + s(age_ce, by = lake, k = 8,
##      m = 1, bs = "tp") + s(lake, bs = "re", k = 8)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.1653      0.1305   1.266   0.206
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(age_ce)      7.733634  9.592 14.187 < 2e-16 ***
## s(age_ce):lakeAzul 3.500538  7.000  4.417 0.01897 *
## s(age_ce):lakeCaldeirao 0.001893  7.000  0.000 0.10059
## s(age_ce):lakeCaveiro 2.250023  7.000  0.743 0.47710
## s(age_ce):lakeEmpadadas Norte 3.657910  7.000 17.377 0.00653 **
## s(age_ce):lakeFunda 4.424174  7.000 11.611 0.00353 **
## s(age_ce):lakeGinjal 1.827819  7.000  0.586 0.09607 .
```

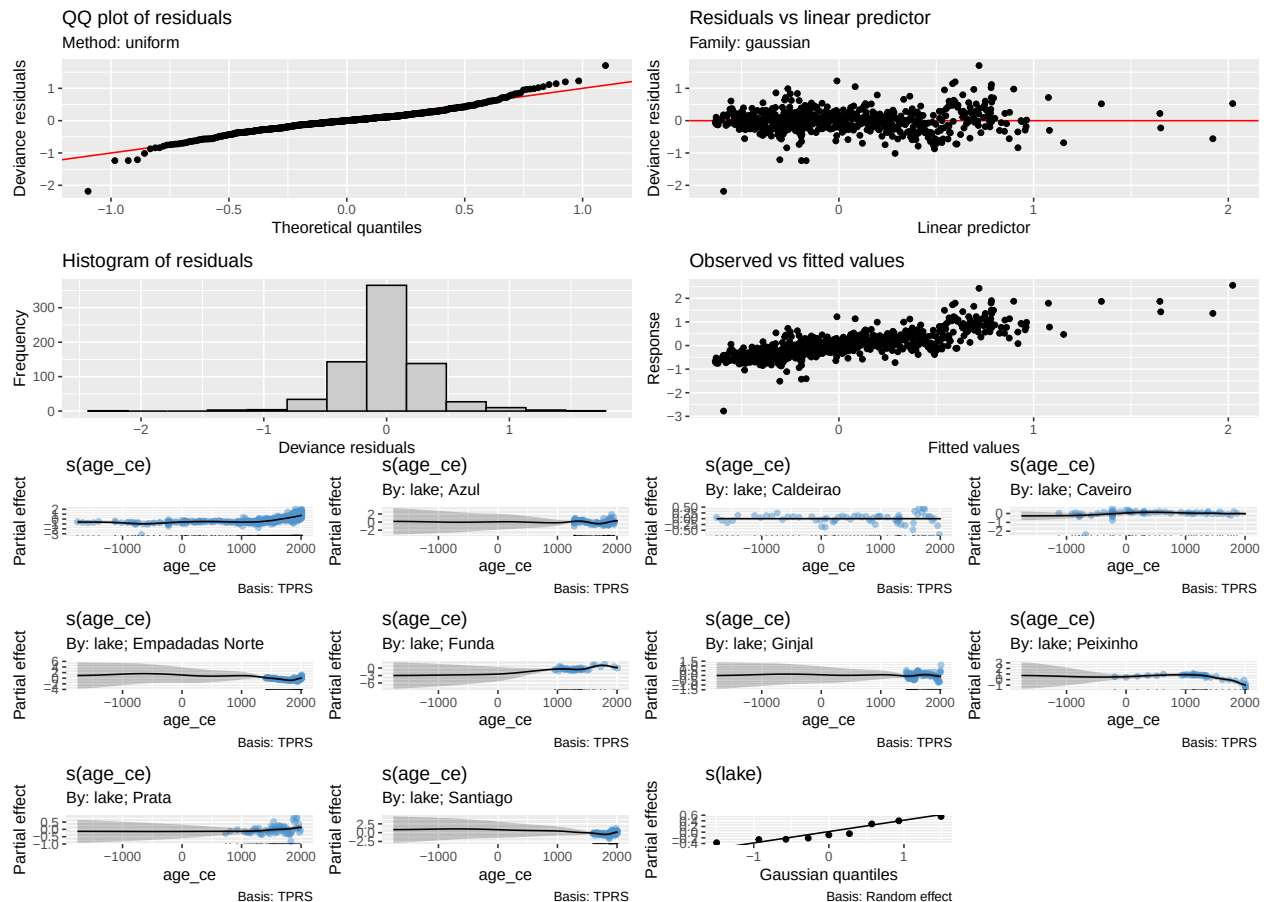
```
## s(age_ce):lakePeixinho      4.694162  7.000 37.474 < 2e-16 ***
## s(age_ce):lakePrata        1.520917  7.000  0.494 0.05522 .
## s(age_ce):lakeSantiago     2.706409  7.000 10.447 0.01364 *
## s(lake)                    6.771457  8.000  8.643 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.609   Deviance explained =  63%
## -REML = 319.76   Scale est. = 0.11751   n = 729

##                               k'      edf    k-index p-value
## s(age_ce)                   29 7.73363422 0.9264204  0.0225
## s(age_ce):lakeAzul          7 3.50053837 0.9264204  0.0150
## s(age_ce):lakeCaldeirao     7 0.00189318 0.9264204  0.0225
## s(age_ce):lakeCaveiro       7 2.25002264 0.9264204  0.0175
## s(age_ce):lakeEmpadadas Norte 7 3.65790965 0.9264204  0.0150
## s(age_ce):lakeFunda         7 4.42417440 0.9264204  0.0300
## s(age_ce):lakeGinjal        7 1.82781867 0.9264204  0.0175
## s(age_ce):lakePeixinho     7 4.69416209 0.9264204  0.0275
## s(age_ce):lakePrata         7 1.52091665 0.9264204  0.0150
## s(age_ce):lakeSantiago     7 2.70640909 0.9264204  0.0175
## s(lake)                     9 6.77145670          NA      NA
```



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 14 iterations.
## Gradient range [-0.0001630094,0.0009169196]
## (score 319.7635 & scale 0.1175101).
## Hessian positive definite, eigenvalue range [3.672284e-05,363.6211].
## Model rank = 102 / 102
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                               k'      edf    k-index p-value
## s(age_ce)                   29.00000  7.73363    0.93  0.010 **
## s(age_ce):lakeAzul          7.00000  3.50054    0.93  0.030 *
## s(age_ce):lakeCaldeirao     7.00000  0.00189    0.93  0.015 *
## s(age_ce):lakeCaveiro       7.00000  2.25002    0.93  0.025 *
## s(age_ce):lakeEmpadadas Norte 7.00000  3.65791    0.93  0.040 *
## s(age_ce):lakeFunda         7.00000  4.42417    0.93  0.020 *
```

```
## s(age_ce):lakeGinjal      7.00000  1.82782    0.93  0.015 *
## s(age_ce):lakePeixinho   7.00000  4.69416    0.93  0.030 *
## s(age_ce):lakePrata      7.00000  1.52092    0.93  0.030 *
## s(age_ce):lakeSantiago   7.00000  2.70641    0.93  0.020 *
## s(lake)                   9.00000  6.77146      NA     NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

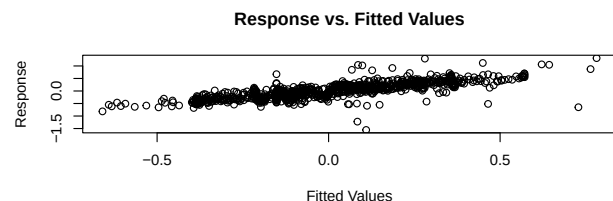
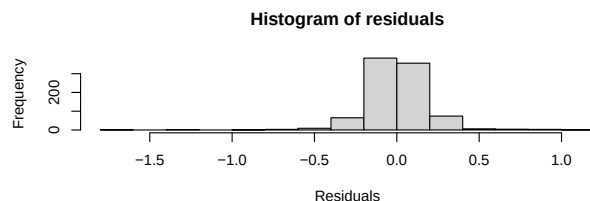
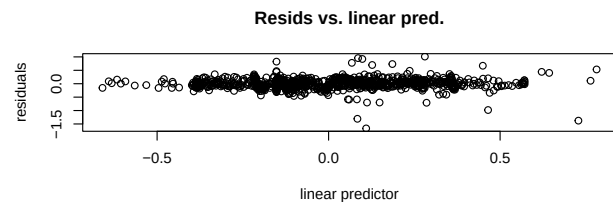
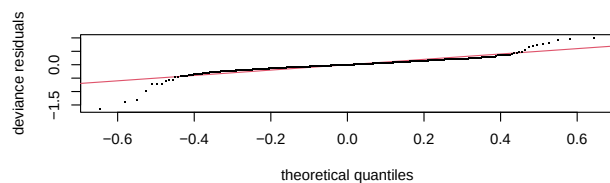


```
## GAM - Fgroups ### Regional Smooth
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## DCA1 ~ s(age_ce, bs = "tp", k = 30) + s(age_ce, by = lake, k = 8,
##      m = 1, bs = "tp") + s(lake, bs = "re", k = 8)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.03841    0.06211   0.618  0.536
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(age_ce)      3.043  3.552  1.655 0.160950
## s(age_ce):lakeAzul  3.642  7.000 11.210 0.000105 ***
## s(age_ce):lakeCaldeirao 4.414  7.000  2.745 1.62e-06 ***
```

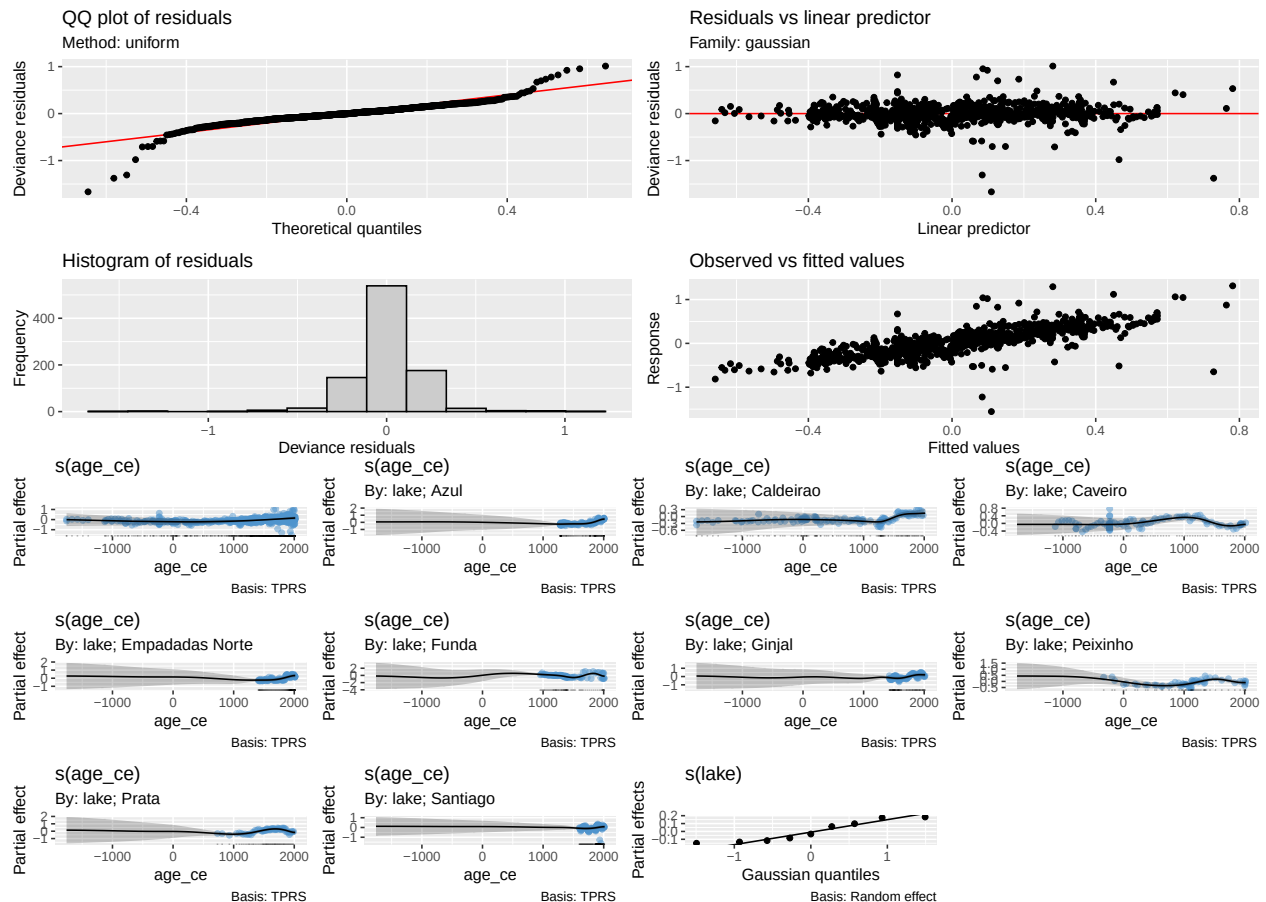
```
## s(age_ce):lakeCaveiro      4.601  7.000  3.101 8.91e-07 ***
## s(age_ce):lakeEmpadadas Norte 3.043  7.000  8.310 5.55e-05 ***
## s(age_ce):lakeFunda       5.132  7.000 21.359 0.005557 **
## s(age_ce):lakeGinjal      2.939  7.000  5.980 0.001999 **
## s(age_ce):lakePeixinho    4.593  7.000  5.075 4.13e-07 ***
## s(age_ce):lakePrata       4.323  7.000  9.235 3.68e-06 ***
## s(age_ce):lakeSantiago    1.807  7.000  1.620 0.002243 **
## s(lake)                   5.790  8.000  4.404 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.612   Deviance explained = 63.1%
## -REML = -107.3   Scale est. = 0.039085   n = 908
```

```
##                               k'      edf k-index p-value
## s(age_ce)                   29 3.042740 0.90268 0.0000
## s(age_ce):lakeAzul          7 3.641540 0.90268 0.0025
## s(age_ce):lakeCaldeirao     7 4.413973 0.90268 0.0050
## s(age_ce):lakeCaveiro       7 4.600862 0.90268 0.0025
## s(age_ce):lakeEmpadadas Norte 7 3.042690 0.90268 0.0025
## s(age_ce):lakeFunda         7 5.132319 0.90268 0.0000
## s(age_ce):lakeGinjal        7 2.939261 0.90268 0.0000
## s(age_ce):lakePeixinho      7 4.592694 0.90268 0.0050
## s(age_ce):lakePrata         7 4.322657 0.90268 0.0075
## s(age_ce):lakeSantiago      7 1.806602 0.90268 0.0000
## s(lake)                     9 5.789803      NA      NA
```



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 11 iterations.
## Gradient range [-1.795511e-07,5.833086e-08]
## (score -107.3023 & scale 0.03908473).
## Hessian positive definite, eigenvalue range [0.2889471,453.0991].
## Model rank = 102 / 102
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                               k'      edf k-index p-value
## s(age_ce)                   29.00  3.04      0.9 <2e-16 ***
## s(age_ce):lakeAzul          7.00  3.64      0.9 <2e-16 ***
```

```
## s(age_ce):lakeCaldeirao      7.00  4.41      0.9 <2e-16 ***
## s(age_ce):lakeCaveiro       7.00  4.60      0.9  0.005 **
## s(age_ce):lakeEmpadadas Norte 7.00  3.04      0.9  0.005 **
## s(age_ce):lakeFundas        7.00  5.13      0.9 <2e-16 ***
## s(age_ce):lakeGinjal        7.00  2.94      0.9 <2e-16 ***
## s(age_ce):lakePeixinho      7.00  4.59      0.9  0.005 **
## s(age_ce):lakePrata         7.00  4.32      0.9  0.005 **
## s(age_ce):lakeSantiago      7.00  1.81      0.9 <2e-16 ***
## s(lake)                     9.00  5.79      NA    NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



Multivariate Analysis - Redundancy Analysis and Variance Partitioning

Data prep - Vegetation change - Regional Vegetation change

```
## Joining with `by = join_by(ID, DCA1)`
## Joining with `by = join_by(ID, DCA1)`
## Joining with `by = join_by(ID, DCA1)`
## Joining with `by = join_by(ID, DCA1)`
## Joining with `by = join_by(ID, DCA1)`
## Joining with `by = join_by(ID, DCA1)`
```

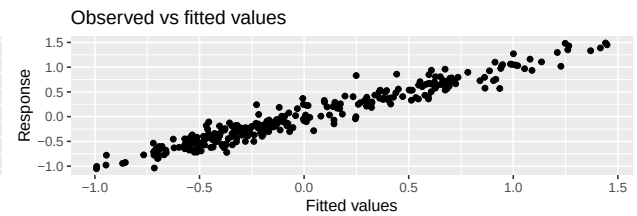
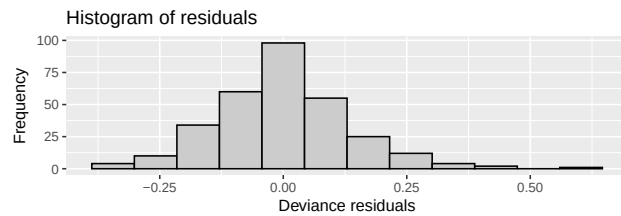
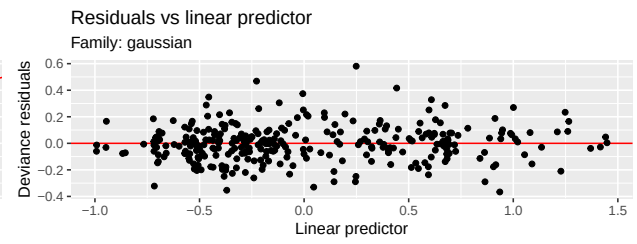
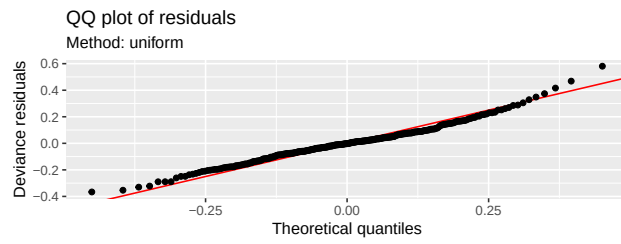
```

## Joining with `by = join_by(ID, DCA1)`

##
## Family: gaussian
## Link function: identity
##
## Formula:
## DCA1 ~ s(age_ce, bs = "tp", k = 25) + s(age_ce, by = sitename,
##      bs = "tp", k = 15, m = 1) + s(sitename, bs = "re")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.04088    0.09552   0.428   0.669
##
## Approximate significance of smooth terms:
##              edf Ref.df      F  p-value
## s(age_ce)      8.288e+00  10.05 39.401 < 2e-16 ***
## s(age_ce):sitenameAzul  5.966e+00  14.00  6.067  0.07317 .
## s(age_ce):sitenameCaldeirão  7.273e-05  14.00  0.000  0.55341
## s(age_ce):sitenameCaldeirão Connor  6.281e+00  13.00 40.323  0.01049 *
## s(age_ce):sitenameCaveiro  7.721e+00  14.00  2.775  0.00223 **
## s(age_ce):sitenameFunda  2.295e+00  14.00  0.381  0.04899 *
## s(age_ce):sitenameGingal  2.841e+00  10.00  1.844  0.21053
## s(age_ce):sitenamePeixinho  9.532e+00  14.00 11.455 < 2e-16 ***
## s(age_ce):sitenamePico Bog  1.001e+01  14.00  8.938  0.00084 ***
## s(age_ce):sitenamePico da Esperança  8.303e+00  14.00  7.099 7.79e-07 ***
## s(sitename)  6.579e+00   8.00  7.765 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.93   Deviance explained = 94.6%
## -REML = -35.381   Scale est. = 0.02352   n = 305

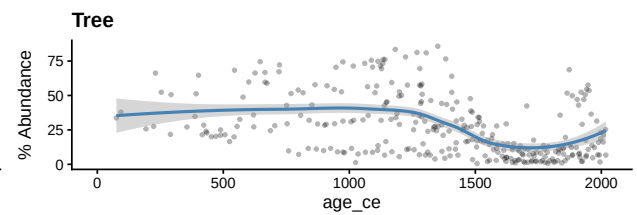
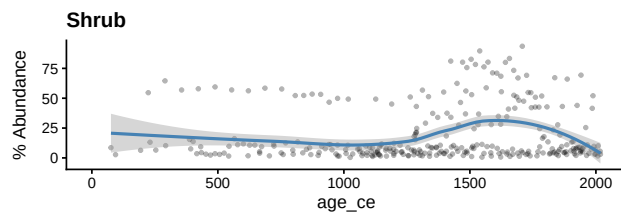
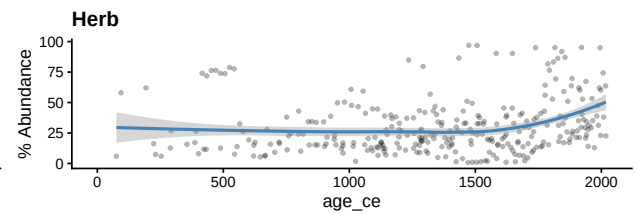
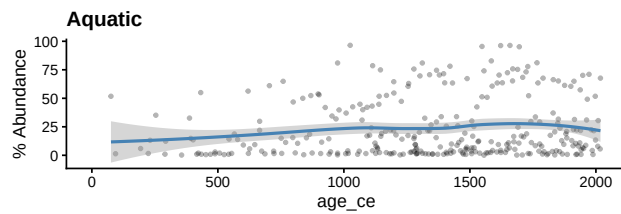
##              k'          edf    k-index p-value
## s(age_ce)      24 8.287646e+00 0.9591675  0.216
## s(age_ce):sitenameAzul  14 5.966097e+00 0.9591675  0.218
## s(age_ce):sitenameCaldeirão  14 7.273118e-05 0.9591675  0.226
## s(age_ce):sitenameCaldeirão Connor  14 6.281298e+00 0.9591675  0.220
## s(age_ce):sitenameCaveiro  14 7.721030e+00 0.9591675  0.230
## s(age_ce):sitenameFunda  14 2.294664e+00 0.9591675  0.226
## s(age_ce):sitenameGingal  14 2.840682e+00 0.9591675  0.210
## s(age_ce):sitenamePeixinho  14 9.532088e+00 0.9591675  0.222
## s(age_ce):sitenamePico Bog  14 1.001306e+01 0.9591675  0.210
## s(age_ce):sitenamePico da Esperança  14 8.302802e+00 0.9591675  0.224
## s(sitename)      9 6.578556e+00      NA      NA

```



```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```

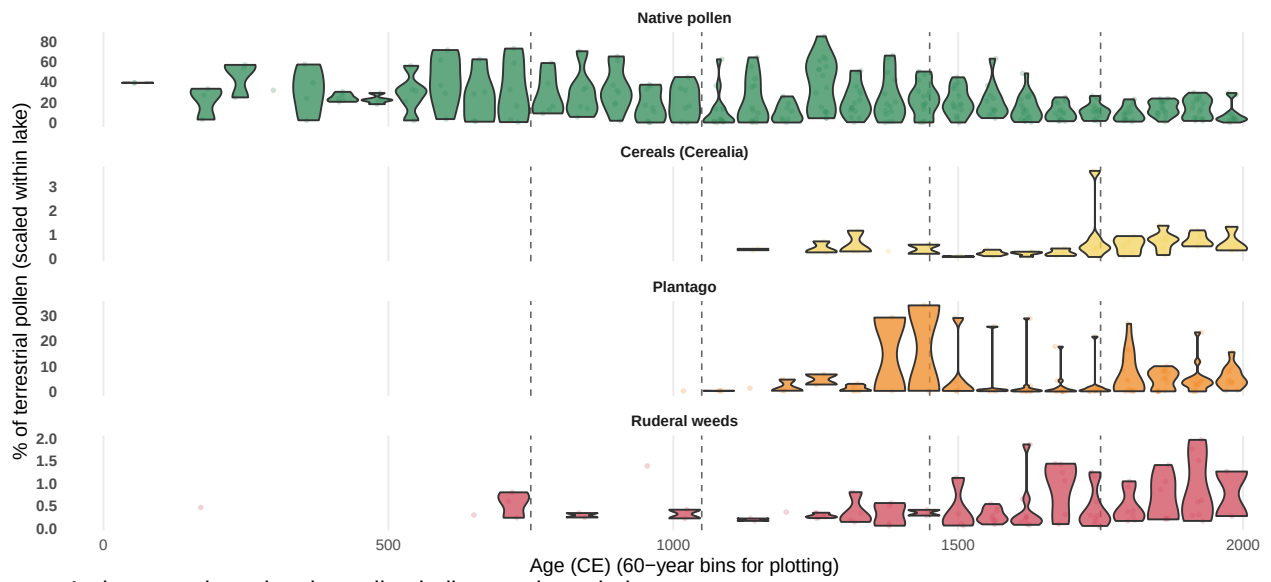
% of Pollen Groups Over Time (GAM fit)



Pollen analysis

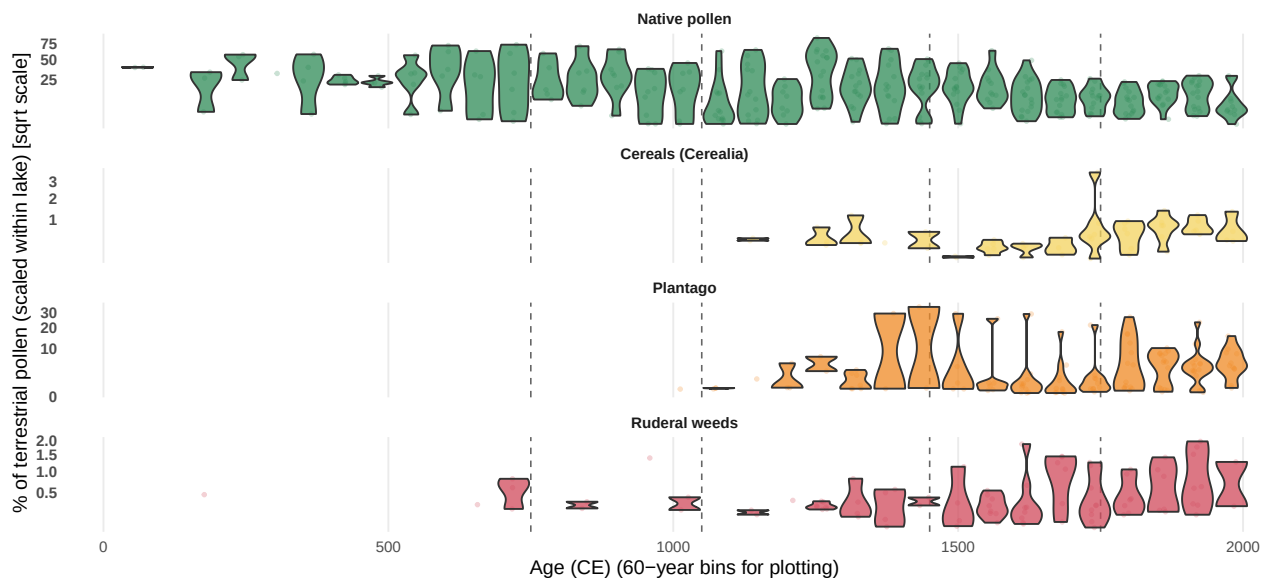
Anthropogenic and native pollen indicators through time

Points show all observations; violins indicate distributions where sufficient data exist

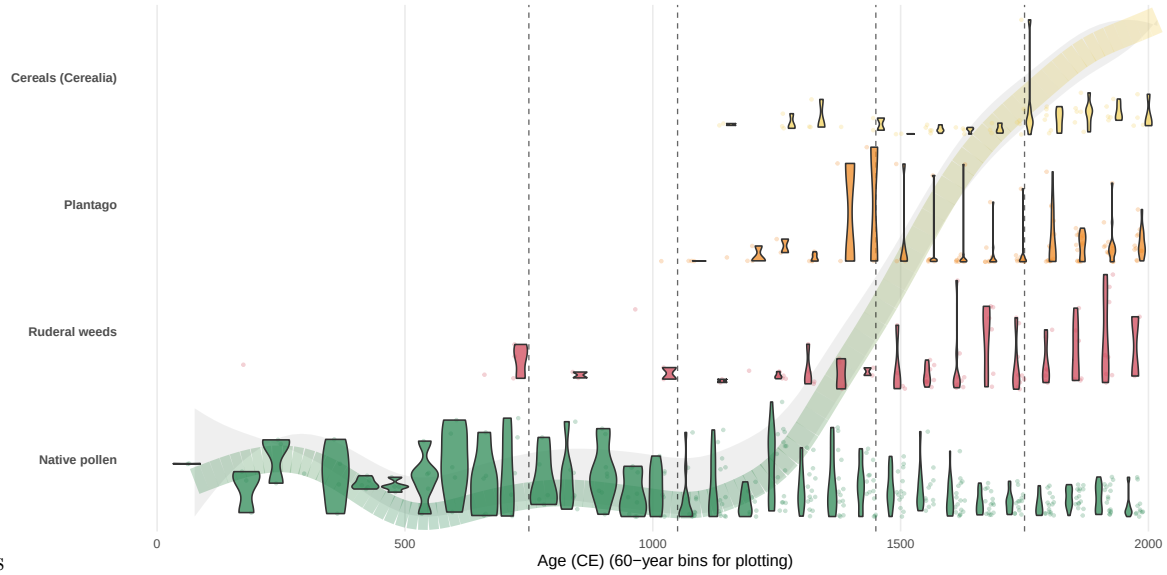


Anthropogenic and native pollen indicators through time

Violin distributions per time bin (across lakes/samples); dashed lines = highlighted dates; sqrt y-scale

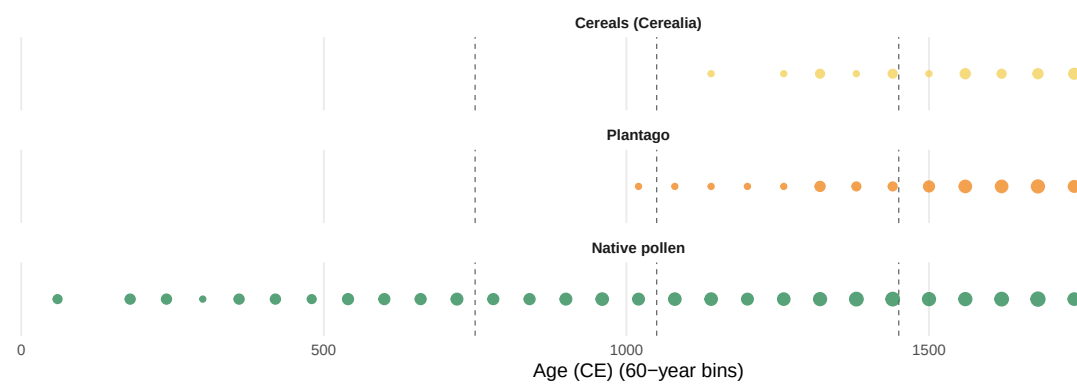


Regional vegetation change with anthropogenic/native pollen indicators
Background: HGAM global vegetation smooth ($\pm$ CI). Overlay: pollen distributions per 60-year bin (scaled within each indicator).



Pollen analysis

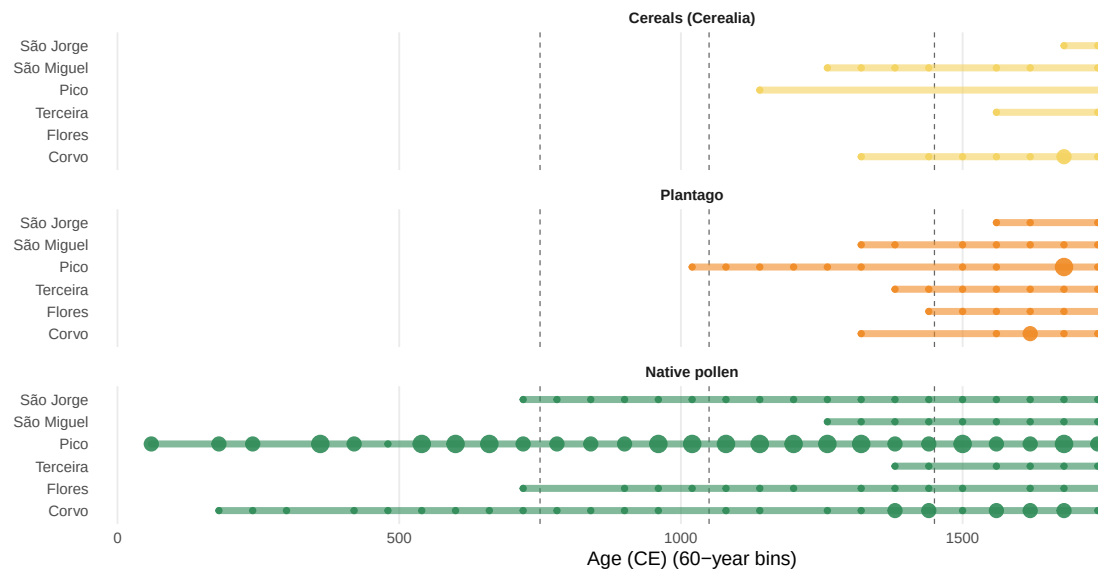
Regional appearance of anthropogenic and native pollen indicators
Each row shows when a pollen group is present in at least one lake; point size reflects number of lakes



Pollen analysis - Regional

Regional appearance of anthropogenic and native pollen indicators

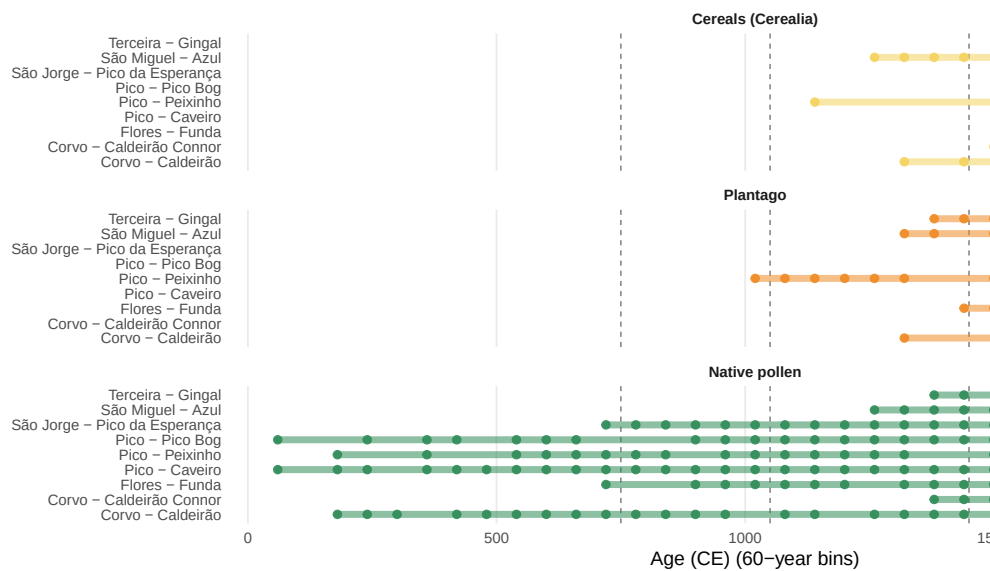
Horizontal bars indicate the temporal span of presence within each island; points show 60-year bins (size = number)



Pollen analysis - by island

Appearance of anthropogenic and native pollen indicators by lake

Horizontal bars show each lake's presence span; points show 60-year bins with presence



Pollen analysis - by island, by lake

Data prep - Align environmental data

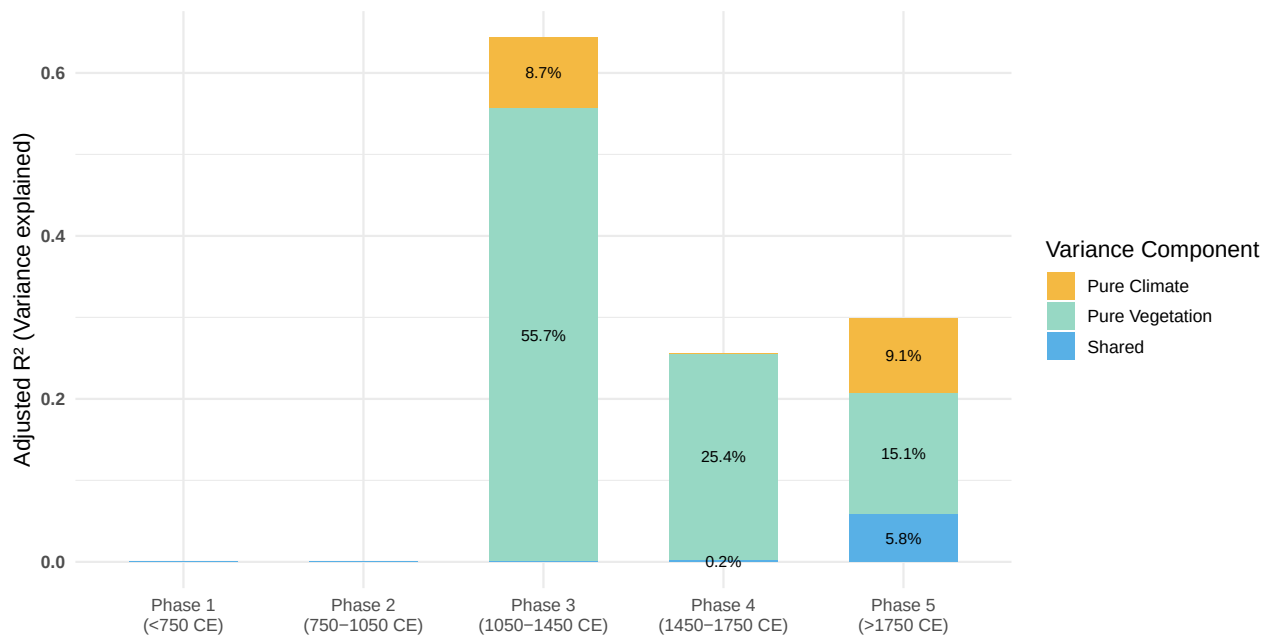
Global RDA model

```
##
## Marginal (single-predictor) tests:
## # A tibble: 3 x 3
##   variable      F      p
##   <chr>      <dbl> <dbl>
## 1 estimate    22.2  0.001
## 2 NAO_Median_Value  3.49  0.012
## 3 Rmean       0.955 0.382
```

```
##
## Final retained model (for biplot + matches varpart):
## Permutation test for capscale under reduced model
## Permutation: free
## Number of permutations: 999
##
## Model: capscale(formula = comm_hel ~ estimate + NAO_Median_Value, data = explanatory)
##           Df SumOfSqs      F Pr(>F)
## Model      2  0.66254 11.597 0.001 ***
## Residual 62  1.77107
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

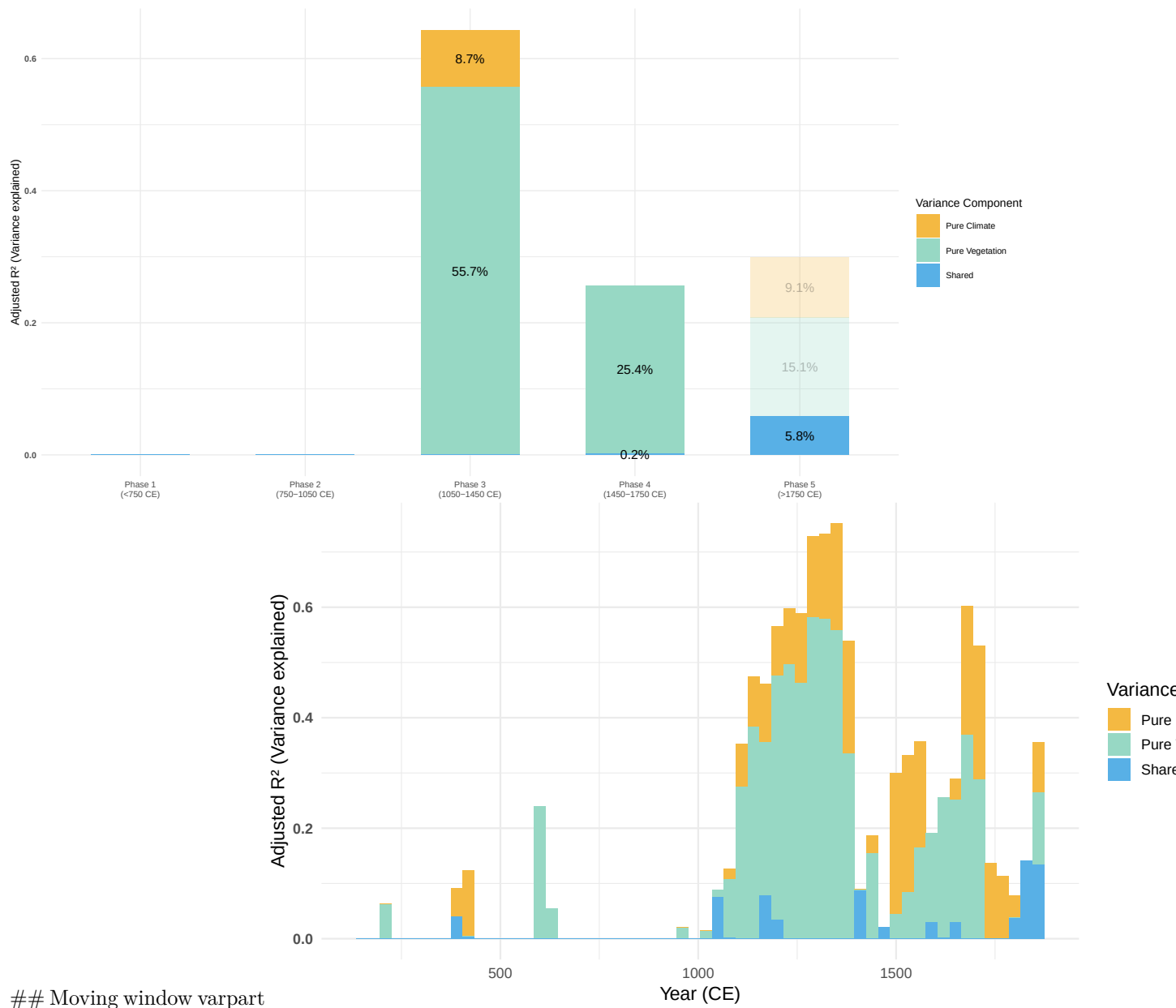
Phase varpart

Phase varpart - Historical phases



Phase varpart - Historical phases - with significances

```
## # A tibble: 5 x 11
##   phase  n_bins mean_n_lakes `Pure NAO_Median_Value` `Pure estimate` Shared
##   <chr>   <int>      <dbl>          <dbl>          <dbl>    <dbl>
## 1 Phase 1     23        1.87            0            0        0
## 2 Phase 2     10        2.8             0            0        0
## 3 Phase 3     14        5.07           0.0867        0.557    0
## 4 Phase 4     10        7.3             0            0.254  0.00240
## 5 Phase 5      8        7.88           0.0912        0.151  0.0578
## # i 5 more variables: p_full <dbl>, p_pure_A <dbl>, p_pure_B <dbl>,
## #   p_seq_A_then_B <dbl>, p_seq_B_then_A <dbl>
```



Moving window varpart

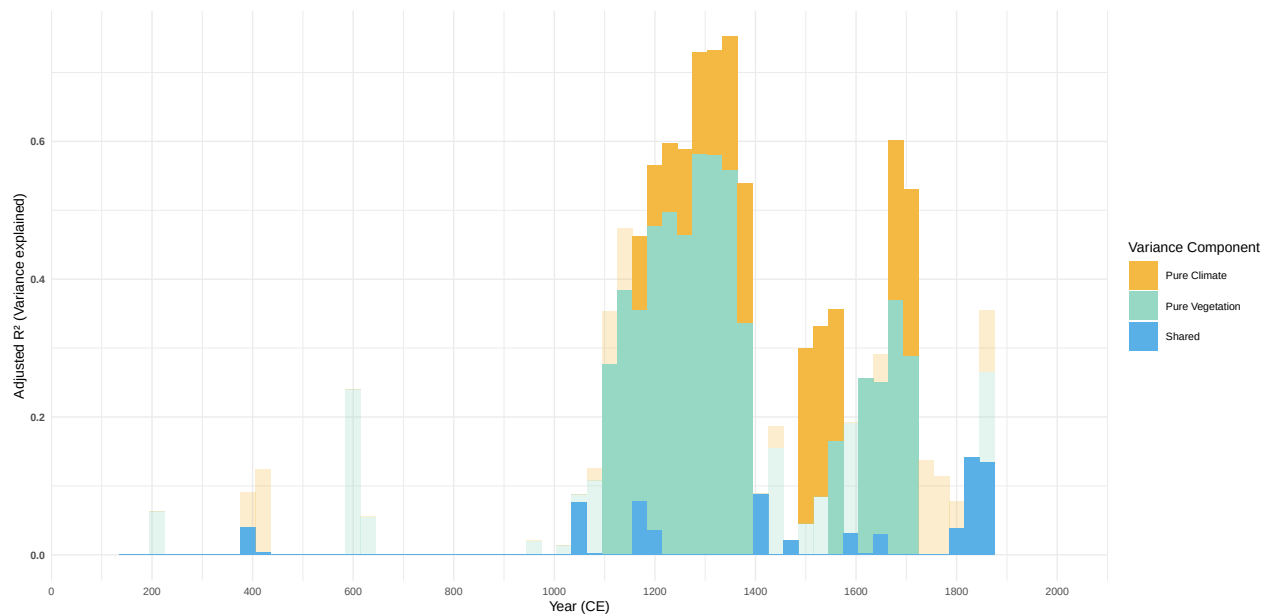
Moving window varpart - with significance

```
## # A tibble: 58 x 13
##   window_start window_end midpoint n_bins mean_n_lakes `Pure NAO_Median_Value`
##   <dbl>         <dbl>     <dbl> <int>     <dbl>         <dbl>
## 1           0         300      150     9         2           0
## 2          30         330      180     9        1.89         0
## 3          60         360      210     9        1.78         0
## 4          90         390      240     9        1.78         0
## 5         120         420      270     9        1.78         0
## 6         150         450      300    10         1.8         0
## 7         180         480      330     9        1.89         0
## 8         210         510      360     9        1.78         0
## 9         240         540      390     9        1.78        0.0513
```

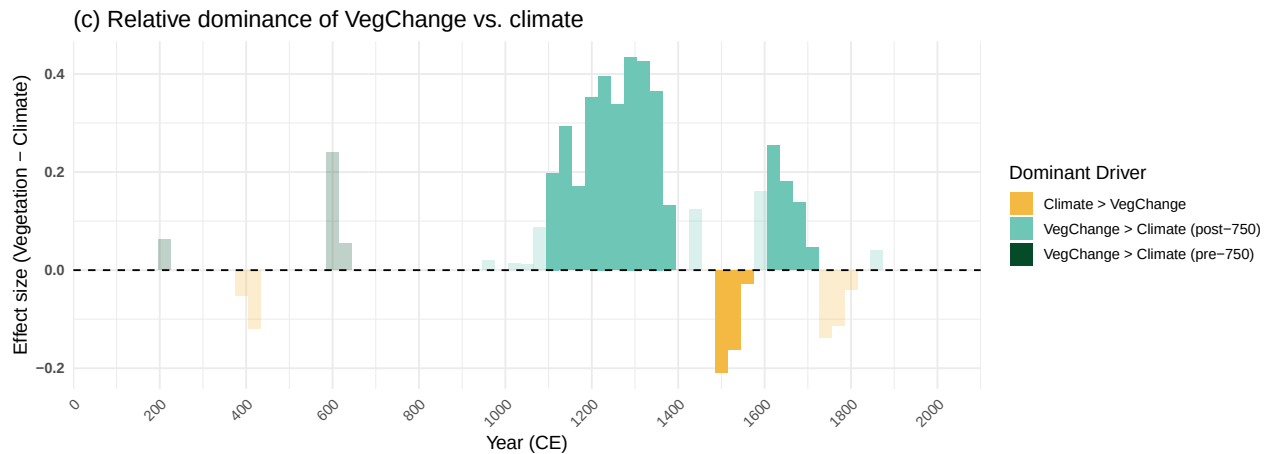
```
## 10          270          570          420          9          2          0.120
## # i 48 more rows
## # i 7 more variables: `Pure estimate` <dbl>, Shared <dbl>, p_full <dbl>,
## #   p_pure_A <dbl>, p_pure_B <dbl>, p_seq_A_then_B <dbl>, p_seq_B_then_A <dbl>

## # A tibble: 58 x 16
##   window_start window_end midpoint n_bins mean_n_lakes `Pure NAO_Median_Value`
##   <dbl>         <dbl>         <dbl> <int>         <dbl>         <dbl>
## 1           0           300          150     9           2           0
## 2          30           330          180     9          1.89          0
## 3          60           360          210     9          1.78          0
## 4          90           390          240     9          1.78          0
## 5         120           420          270     9          1.78          0
## 6         150           450          300    10           1.8           0
## 7         180           480          330     9          1.89          0
## 8         210           510          360     9          1.78          0
## 9         240           540          390     9          1.78          0.0513
## 10        270           570          420     9           2          0.120
## # i 48 more rows
## # i 10 more variables: `Pure estimate` <dbl>, Shared <dbl>, p_full <dbl>,
## #   p_pure_A <dbl>, p_pure_B <dbl>, p_seq_A_then_B <dbl>, p_seq_B_then_A <dbl>,
## #   full_sig <chr>, pure_A_sig <chr>, pure_B_sig <chr>

## # A tibble: 1 x 4
##   n_windows n_full_sig n_pure_A_sig n_pure_B_sig
##   <int>     <int>     <int>     <int>
## 1       58        18        13        15
```



Effect sizes with significances



Sensitivity analyses

Sensitivity analyses - PHASE-BOUNDARY ROBUSTNESS (GRID TEST)

```
## # A tibble: 15 x 5
##   phase component      n_runs max_abs_delta median_abs_delta
##   <chr>   <chr>      <int>      <dbl>      <dbl>
## 1 Phase 1 Pure NAO_Median_Value      625          0          0
## 2 Phase 1 Pure estimate      625      0.0122          0
## 3 Phase 1 Shared      625      0.0116          0
## 4 Phase 2 Pure NAO_Median_Value      625          0          0
## 5 Phase 2 Pure estimate      625      0.0201          0
## 6 Phase 2 Shared      625          0          0
## 7 Phase 3 Pure NAO_Median_Value      625      0.0388      0.0112
## 8 Phase 3 Pure estimate      625      0.0972      0.0464
## 9 Phase 3 Shared      625          0          0
## 10 Phase 4 Pure NAO_Median_Value      625      0.439      0.0462
## 11 Phase 4 Pure estimate      625      0.197      0.0681
## 12 Phase 4 Shared      625      0.0740      0.00240
## 13 Phase 5 Pure NAO_Median_Value      625      0.0912      0.0298
## 14 Phase 5 Pure estimate      625      0.0958      0.0689
## 15 Phase 5 Shared      625      0.170      0.0766
```

```
## # A tibble: 5 x 2
##   phase prop_runs_B_dominant
##   <chr>      <dbl>
## 1 Phase 1          1
## 2 Phase 2          1
## 3 Phase 3          1
## 4 Phase 4        0.92
## 5 Phase 5          1
```

Sensitivity analyses - Leave-one-out per phases

```
## # A tibble: 9 x 14
##   leave_out mean_pure_A mean_pure_B mean_shared median_pure_A median_pure_B
##   <chr>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 Funda      0.0249      0.249  0.0536          0      0.154
```

```
## 2 Prata          0.0482      0.161 0.0137          0.00565      0.128
## 3 Ginjal         0.0459      0.164 0.00511          0          0.127
## 4 Azul           0.0203      0.167 0.00000383        0          0.171
## 5 Caldeirao      0.0452      0.205 0.0168          0.00397      0.141
## 6 Santiago       0.0381      0.180 0.00379          0.000278     0.122
## 7 Empadadas Nor~ 0.0344      0.202 0.00168          0          0.125
## 8 Peixinho       0.0256      0.191 0.0219          0.0168       0.151
## 9 Caveiro        0.0206      0.193 0.00665          0          0.173
## # i 8 more variables: median_shared <dbl>, mean_diff_BA <dbl>,
## #   delta_mean_pure_A <dbl>, delta_mean_pure_B <dbl>, delta_mean_shared <dbl>,
## #   n_phases <int>, prop_phases_B_dominant <dbl>, prop_phases_A_dominant <dbl>

## # A tibble: 9 x 16
##   leave_out      Phase 1 - Pure NAO_M~1 Phase 1 - Pure estim~2 `Phase 1 - Shared`
##   <chr>          <dbl>          <dbl>          <dbl>
## 1 Azul          0          0          0
## 2 Caldeirao     0          0          0
## 3 Caveiro       0          0          0
## 4 Empadadas No~ 0          0          0
## 5 Funda         0          0          0
## 6 Ginjal        0          0          0
## 7 Peixinho      0          0          0
## 8 Prata         0          0.0122       0
## 9 Santiago      0          0          0
## # i abbreviated names: 1: `Phase 1 - Pure NAO_Median_Value`,
## #   2: `Phase 1 - Pure estimate`
## # i 12 more variables: `Phase 2 - Pure NAO_Median_Value` <dbl>,
## #   `Phase 2 - Pure estimate` <dbl>, `Phase 2 - Shared` <dbl>,
## #   `Phase 3 - Pure NAO_Median_Value` <dbl>, `Phase 3 - Pure estimate` <dbl>,
## #   `Phase 3 - Shared` <dbl>, `Phase 4 - Pure NAO_Median_Value` <dbl>,
## #   `Phase 4 - Pure estimate` <dbl>, `Phase 4 - Shared` <dbl>, ...
```

Sensitivity analyses - Leave one out - moving window

```
## # A tibble: 9 x 9
##   leave_out      mean_pure_A mean_pure_B median_pure_A median_pure_B mean_diff_BA
##   <chr>          <dbl>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 Azul          0.0382      0.108      0          0          0.0695
## 2 Caldeirao     0.0835      0.131     0.00199     0.00420     0.0480
## 3 Caveiro       0.0562      0.120      0          0.0173     0.0643
## 4 Empadadas No~ 0.0508      0.124      0          0.00416     0.0733
## 5 Funda         0.0525      0.144      0          0.0198     0.0915
## 6 Ginjal        0.0422      0.111      0          0.00732     0.0686
## 7 Peixinho      0.0476      0.108      0          0          0.0602
## 8 Prata         0.0762      0.102     0.0183     0.00790     0.0258
## 9 Santiago      0.0551      0.117      0          0.000966    0.0621
## # i 3 more variables: n_windows <int>, prop_windows_B_dominant <dbl>,
## #   prop_windows_A_dominant <dbl>

## # A tibble: 9 x 11
##   leave_out      mean_pure_A mean_pure_B median_pure_A median_pure_B mean_diff_BA
##   <chr>          <dbl>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 Azul          0.0382      0.108      0          0          0.0695
## 2 Caldeirao     0.0835      0.131     0.00199     0.00420     0.0480
## 3 Caveiro       0.0562      0.120      0          0.0173     0.0643
```



```
## 4 Empadadas No~      0.0508      0.124      0      0.00416      0.0733
## 5 Funda              0.0525      0.144      0      0.0198      0.0915
## 6 Ginjal             0.0422      0.111      0      0.00732     0.0686
## 7 Peixinho          0.0476      0.108      0      0          0.0602
## 8 Prata              0.0762      0.102     0.0183     0.00790     0.0258
## 9 Santiago           0.0551      0.117      0      0.000966    0.0621
## # i 5 more variables: n_windows <int>, prop_windows_B_dominant <dbl>,
## #   prop_windows_A_dominant <dbl>, delta_mean_pure_A <dbl>,
## #   delta_mean_pure_B <dbl>
```

Sensitivity analyses - lake record position / length

FIGURES

FIGURE 2a: NAO panel on top + main panel (DCA + vegetation)

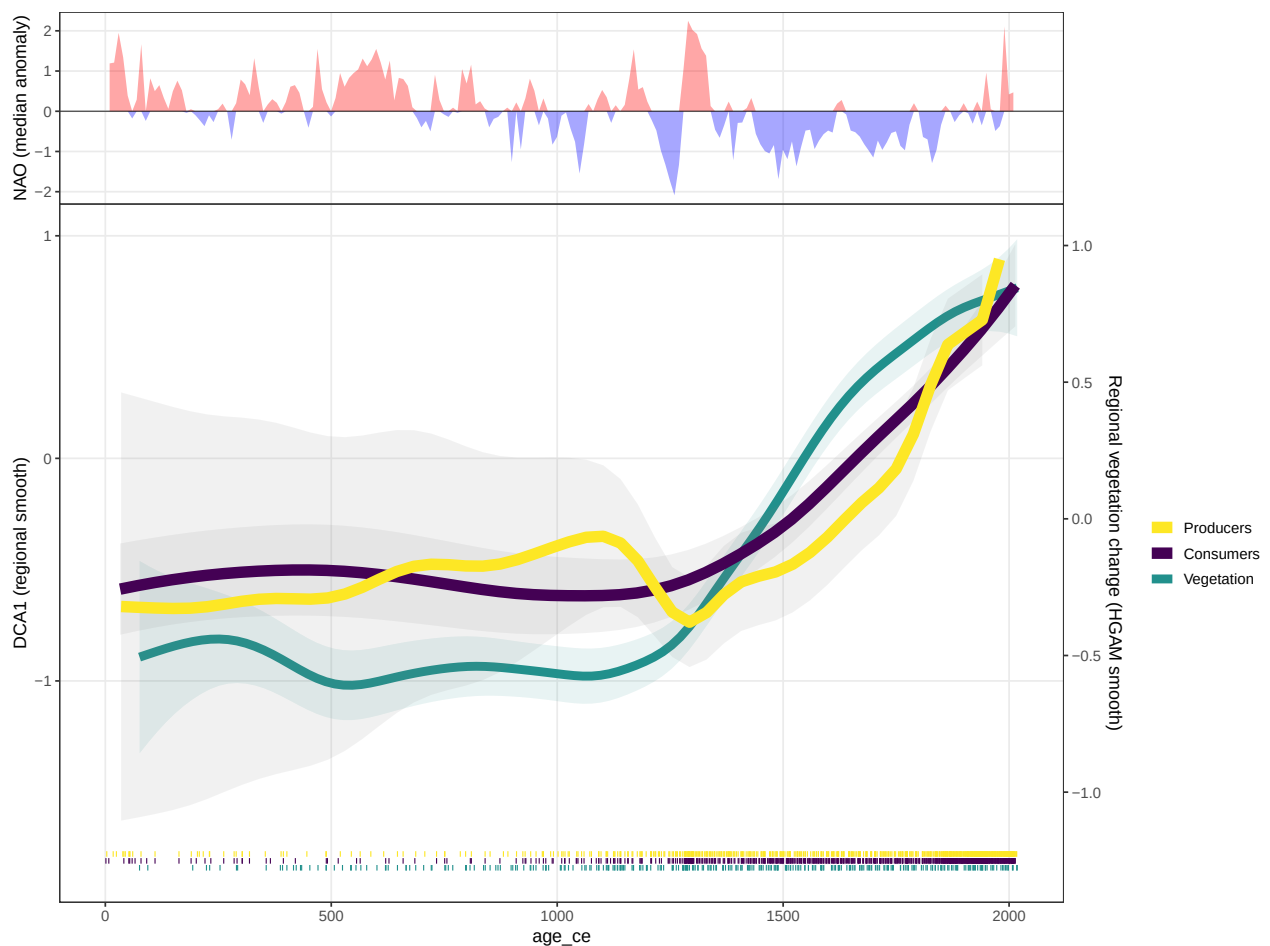


FIGURE 2b - Individual lake DCAs plus map

```
## `geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
```

```
## `geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
## Coordinate system already present.
## i Adding new coordinate system, which will replace the existing one.
```

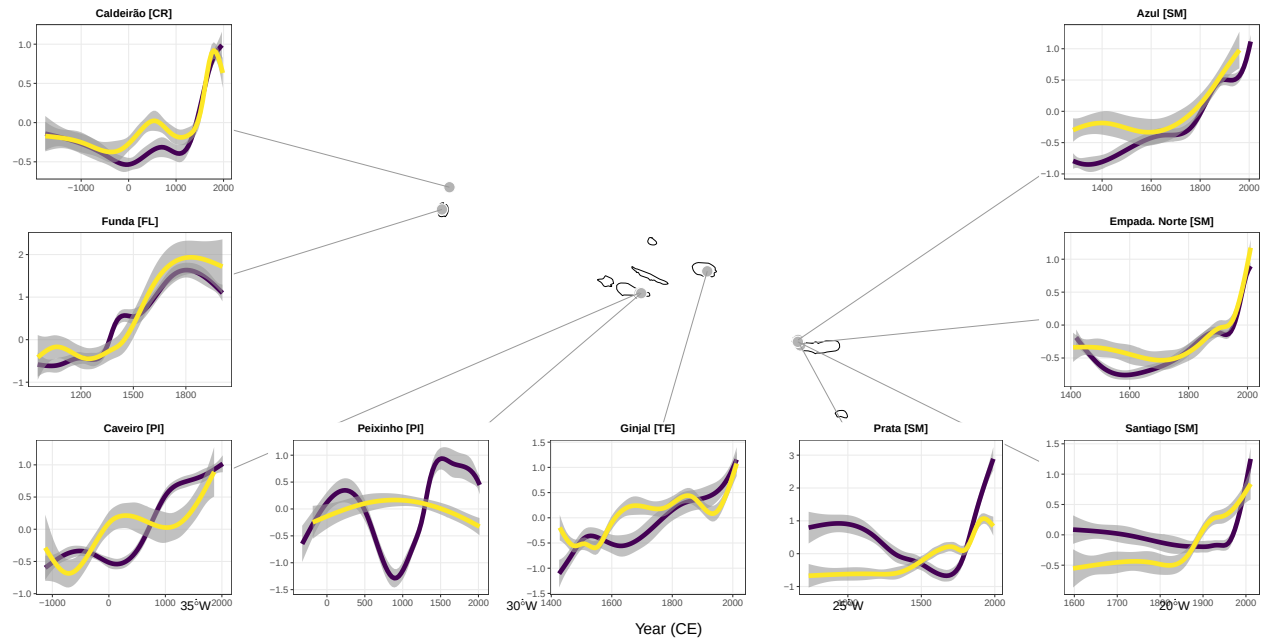


FIGURE 3 - Trophic structure classification

```
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```

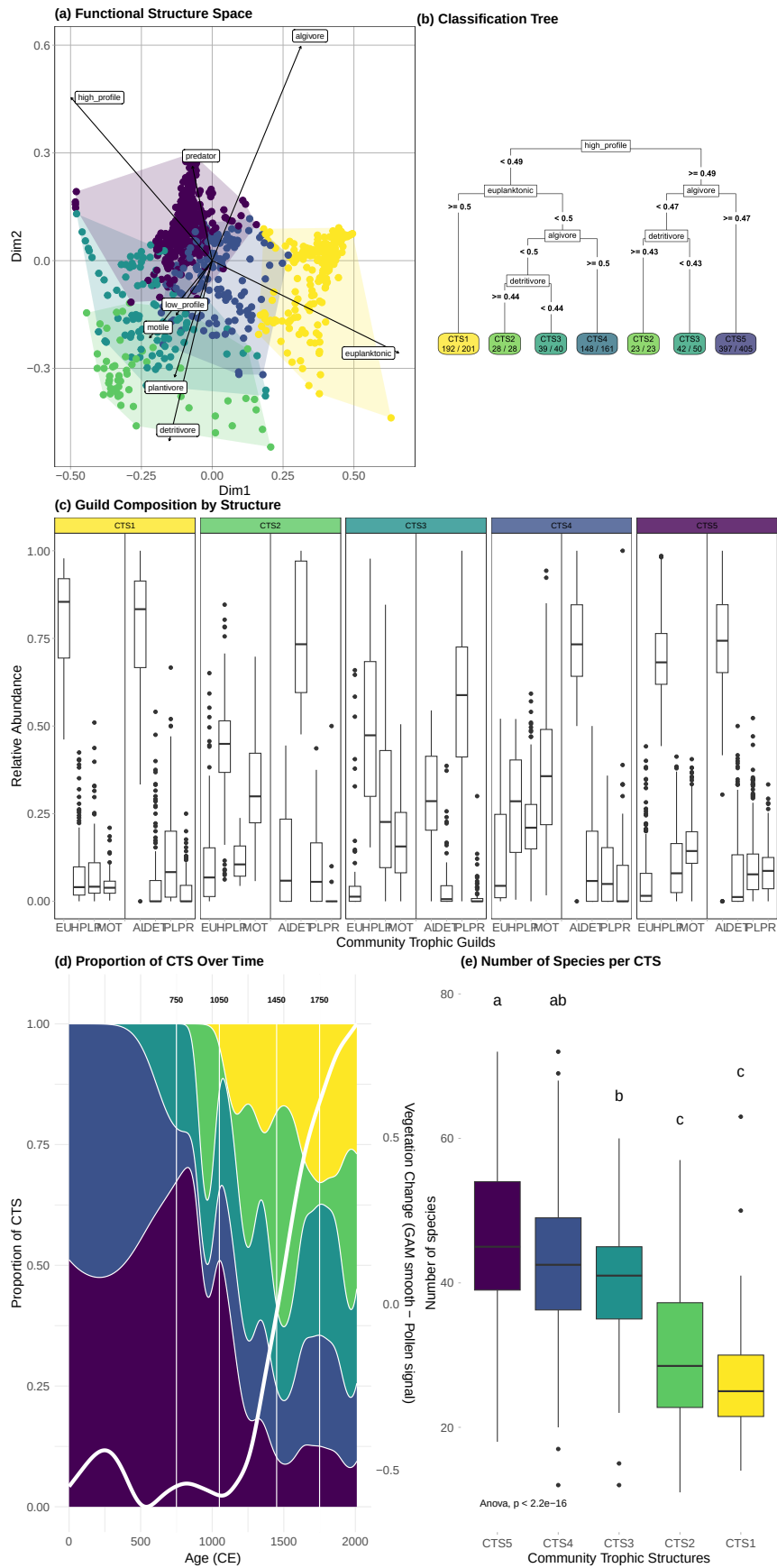
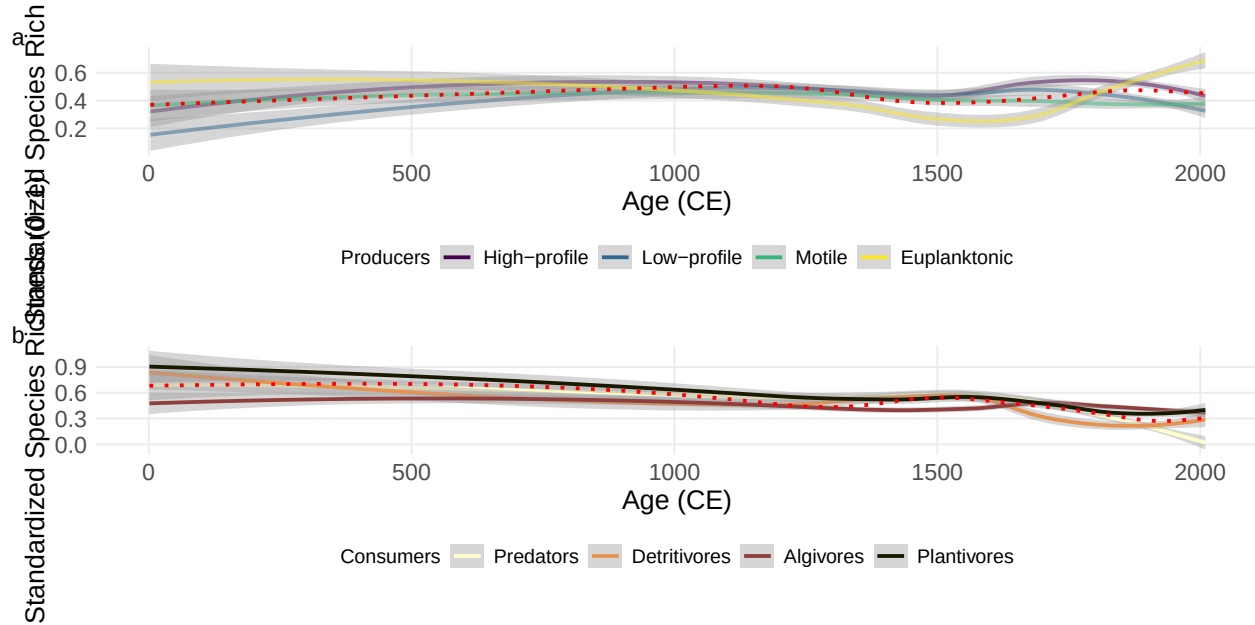


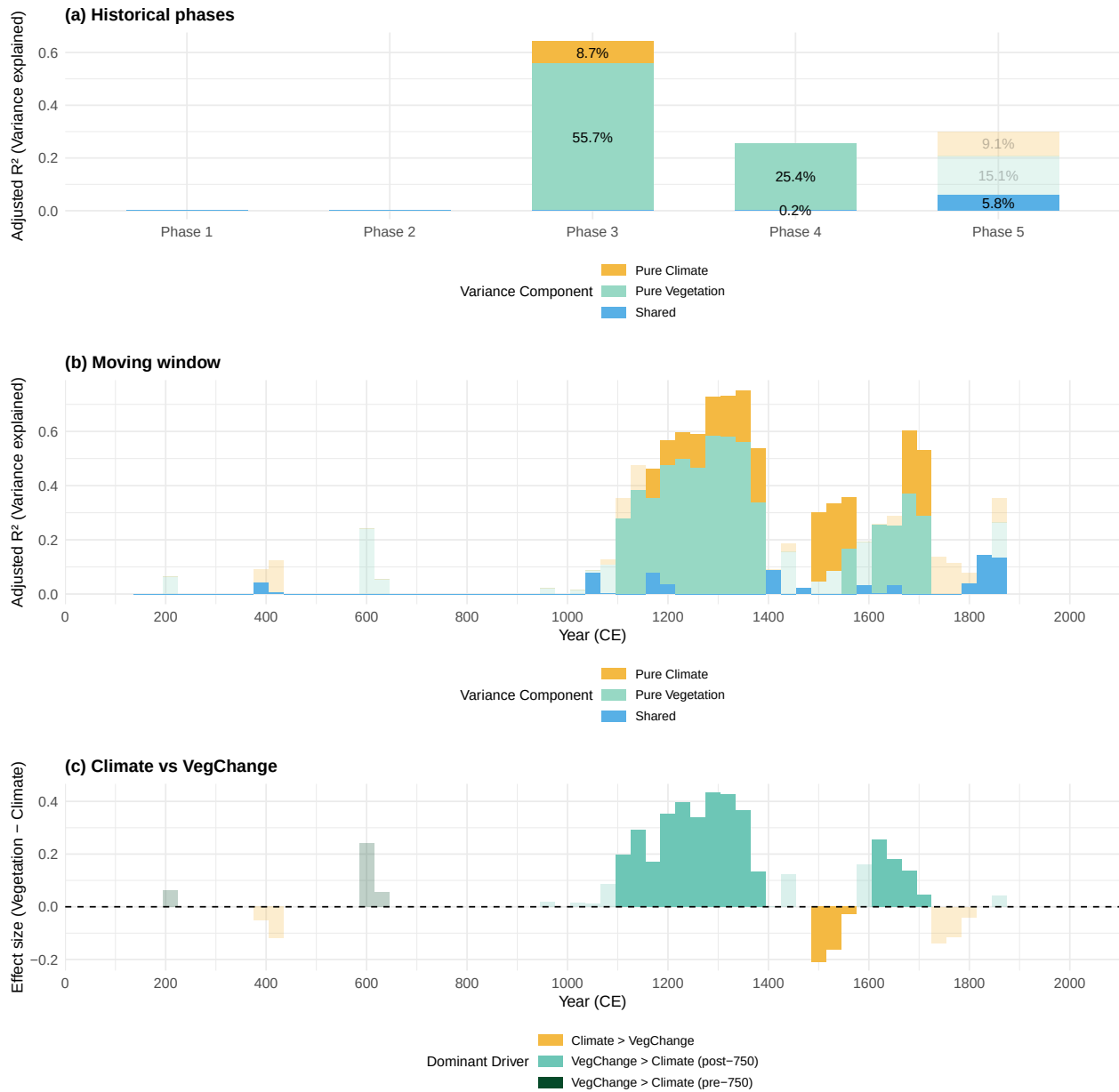
FIGURE 4 - Diversity Dynamics by Functional Group (Standardized Species Richness)

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

FIGURE 5 - Historical and Moving Window Variance Partitioning



SUPPLEMENTARY FIGURES

SUPPLEMENTARY FIGURE S2 - Climatic reconstructions

pdf

2

SUPPLEMENTARY FIGURE S3 - Pollen GAM

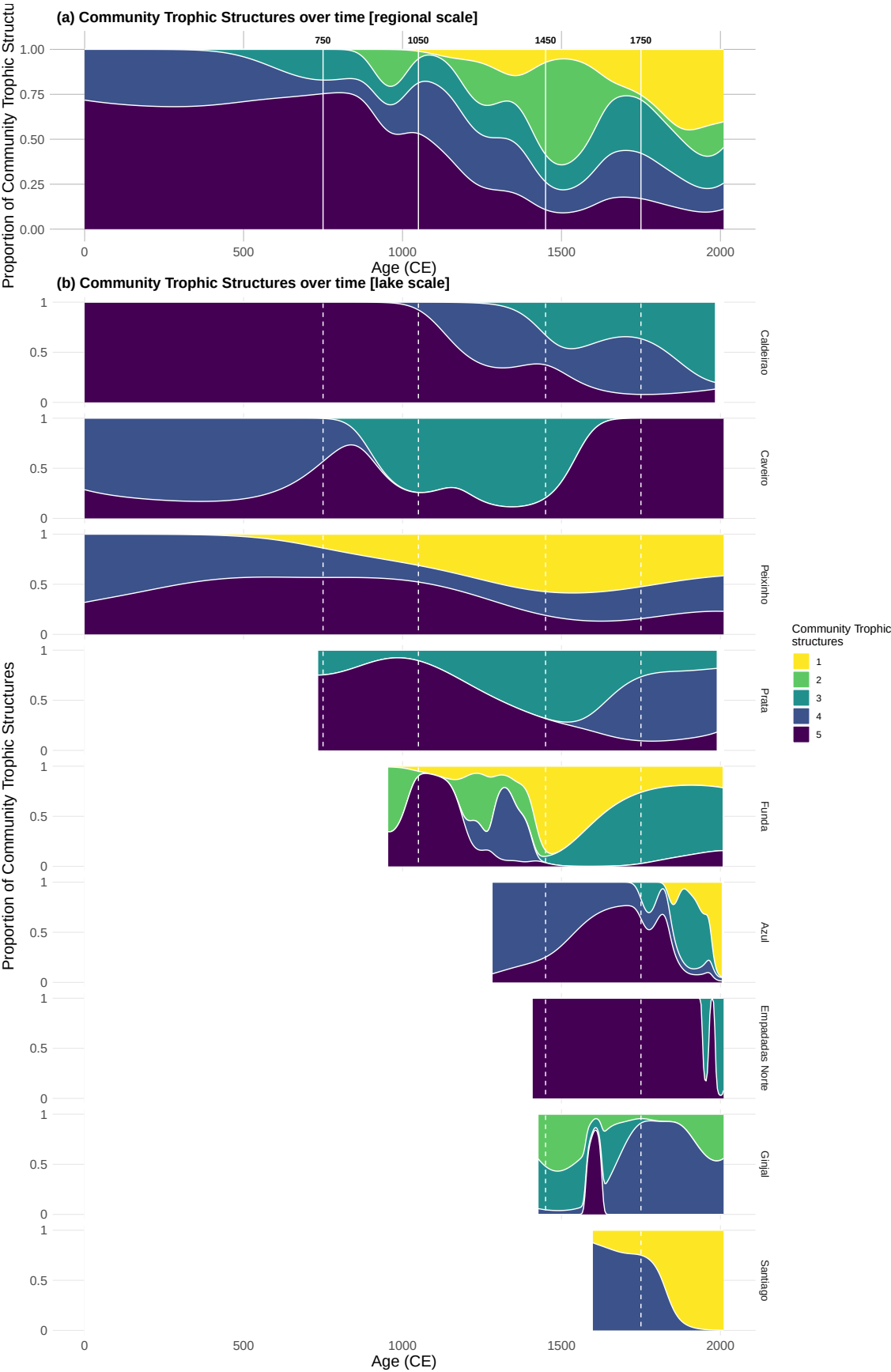
pdf

2

SUPPLEMENTARY FIGURE S4 - Pollen GAM by island

```
## pdf
## 2
```

SUPPLEMENTARY FIGURE S5 - Community Trophic Structures over time



SUPPLEMENTARY FIGURE S6 - Variable Importance Plot

```
## pdf  
## 2
```

SUPPLEMENTARY FIGURE S7 - Global RDA

```
## pdf  
## 2
```

SUPPLEMENTARY FIGURE S9 - Null model

```
## pdf  
## 2
```